

Stress-Risk Detection in a Persian Pregnancy Forum

Data Construction, Active Learning, Transformer-Tabular Fusion, Explainability, and Human-in-the-Loop Monitoring

Course	APPLIED DATA SCIENCE
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Report status	Full-project report; Member 1, Member 2, and Member C modeling sections integrated
Date	August 2026

Scope and safety statement

This project estimates a research stress-risk score from public forum posts. It is not a medical diagnosis, does not replace validated clinical screening, and must not autonomously determine emergency or medical action. The intended operational interpretation is a private human-in-the-loop prioritization overlay.

Abstract

Pregnancy-related online forums contain large volumes of spontaneous discussion about physical symptoms, family conflict, loss, fear, financial pressure, and emotional distress. Yet Persian-language resources for computational analysis of perinatal stress are scarce, and no ready-made labeled dataset existed for the Ninisite forum used in this project. We therefore built an end-to-end pipeline from raw website acquisition to a reproducible stress-risk model. The work included robust scraping, raw-text-preserving cleaning with local Persian normalization, structured feature engineering, exploratory stress-proxy construction, AI-assisted multi-dimensional annotation, active-learning expansion, duplicate-aware label reconstruction, leakage-safe group splitting, five-fold out-of-fold prediction generation, and learned late fusion of a tabular CatBoost branch with a Persian ParsBERT text branch. The structured Member 1 branch used a weighted CatBoost regressor over 67 ordered post, user, temporal, categorical, signature, and lexical-count features and generated leakage-safe training OOF predictions plus five-fold ensemble predictions for validation and test. The final system was trained as continuous stress regression and mapped to four operational bins: Low, Moderate, High, and Very High. The selected hybrid CatBoost fusion achieved a locked-test mean absolute error of 0.9174, outperforming the same-test tabular branch (1.2889) and Transformer branch (0.9689). On the locked test, recall was 87.54% for Low, 41.79% for Moderate, 52.38% for High, and 76.92% for Very High. Three of four internal recall objectives were met; Moderate remained the primary improvement area. The report presents the engineering challenges, labeling and active-learning design, leakage controls, model comparisons, explainability analysis, limitations, ethical requirements, and a proposed human-in-the-loop deployment pathway based on repeated and temporally aggregated risk rather than autonomous diagnosis.

Keywords: Persian NLP; pregnancy; perinatal mental health; stress detection; online forum; active learning; ParsBERT; CatBoost; late fusion; SHAP; human-in-the-loop.

Team contributions note

Member 1: structured feature engineering and CatBoost tabular modeling.

Member 2: Persian text preprocessing, ParsBERT training, and leakage-safe OOF/ensemble predictions.

Member C: scraping, cleaning, canonical data foundation, grouped split design, late fusion, calibration,

Shared work: annotation design, active learning, interpretation, and report review.

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1. Problem Definition and Research Objectives

1.1 Problem statement

The project addresses automated estimation of stress-risk from Persian user-generated content in the topic and discussion areas of Ninisite.com, a large pregnancy and family forum. The central challenge is not ordinary sentiment analysis. A post can be negative without expressing severe stress, and a short neutral-looking message can occur inside a longer crisis trajectory. The project therefore models stress as a continuous score, uses operational risk bins for evaluation and monitoring, and preserves the ability to aggregate multiple posts over time.

The project started without a ready-made research dataset, clinical labels, clean identifiers, or an established Persian benchmark for this exact domain. Consequently, data acquisition, cleaning, exploratory sampling, label creation, split design, model development, and reproducibility were all part of the research contribution rather than preliminary clerical work.

1.2 Operational target

Operational class	Authoritative true-score definition	Project interpretation
Low	$\text{stress} \leq 3$	Little or no stress evidence in the annotated post
Moderate	$3 < \text{stress} \leq 5$	Meaningful but not clearly severe stress
High	$5 < \text{stress} \leq 7$	Strong stress evidence requiring attention in context
Very High	$\text{stress} > 7$	Potentially severe or dangerous distress signal for rapid human review

Important distinction

The four labels are operational bins derived from a continuous project score. They are not clinical diagnoses and must not be presented as validated psychiatric categories.

1.3 Research questions

- Can a continuous stress-risk score be learned from Persian pregnancy-forum posts and associated metadata?
- Does combining a tabular model with a Persian Transformer improve over either branch alone or over fixed averaging?
- Can active learning expand the scarce high-stress region more efficiently than random labeling?
- Can leakage-safe grouped splits and OOF stacking provide credible validation and locked-test estimates?
- Can explainability and temporal aggregation support a responsible human-review workflow?

1.4 Main project contributions

- A robust scraping and recovery workflow for a large, unstable Persian forum.
- A cleaned website-scale dataset of approximately 360,000 posts with text, metadata, temporal, behavioral, and lexical features.
- A data-driven differential Persian positive/negative lexicon and exploratory stress proxy used for sampling rather than as final ground truth.
- A canonical continuous-stress label set reconstructed from multiple annotation rounds with provenance, calibration, and duplicate resolution.
- Author/thread/content-connected leakage controls, four final data roles, and five grouped OOF folds.
- A Persian text branch, a tabular branch, and a learned hybrid fusion selected under project-defined safety constraints.
- A locked-test evaluation, SHAP-based explanation, temporal alert prototype, and reproducible notebook projects.

2. Importance and Potential Impact

Perinatal mental health is a material public-health concern. The World Health Organization reports that more than 10% of pregnant women and women who have recently given birth experience depression, while major barriers to care include limited investment, shortages of trained providers, and stigma [1]. Stressful life events during pregnancy have also been associated in meta-analysis with increased risks of preterm birth, low birth weight, and small-for-gestational-age outcomes [2]. These findings do not imply that forum language can diagnose these outcomes; they establish why early attention to distress may matter.

Access to perinatal mental-health assessment and treatment is uneven. Systematic reviews identify individual, organizational, social, cultural, and structural barriers, including stigma, fragmented services, lack of awareness, language and cultural barriers, unclear referral pathways, and limited resources [3]. Iranian studies likewise report substantial psychological distress, anxiety, or depression during pregnancy and postpartum [8,9].

Online forums are particularly relevant because users may disclose experiences there that they do not immediately communicate to professionals. Research on perinatal forums suggests that forum use can interact with stigma and disclosure, and that online communities may support help-seeking for some users [4]. The project team also observed that Ninisite contains unstructured narratives from users with varied access to information and services. The current dataset, however, does not prove the website's demographic representativeness; claims about rural residence or education should be validated before publication.

Potential benefit - not a guarantee

A well-governed system could help moderators or qualified support staff notice repeated severe distress earlier. It cannot guarantee prevention of harm, and it should never contact emergency or medical organizations automatically without an approved human, privacy, consent, and legal protocol.

3. Related Work and Project Novelty

3.1 Mental-health detection from social media

Machine-learning research has repeatedly shown that language and behavioral signals from social media can support research-level detection of depression, anxiety, and psychological distress. A 2022 scoping review included 54 studies across platforms such as Twitter, Facebook, Instagram, Reddit, and Sina Weibo, while emphasizing that such systems should complement rather than replace traditional screening [6]. NLP has also been used to estimate population-level distress trends from public social-media text [7].

Pregnancy-specific work is smaller but directly relevant. Dhankar and Katz developed a Transformer-enabled pipeline for estimating depression and general anxiety in pregnancy-related Reddit communities and used topic modeling to characterize concerns [5]. Moore, Drey, and Ayers studied relationships among perinatal forum use, stigma, and disclosure to health-care providers [4]. These studies support the value of online narratives but use different platforms, languages, labels, and evaluation designs.

3.2 Persian NLP and modeling foundations

BERT introduced bidirectional Transformer pretraining that can be fine-tuned for downstream language tasks [10]. ParsBERT adapted this paradigm to Persian and reported improvements over multilingual and previous Persian baselines on tasks such as sentiment analysis, text classification, and named-entity recognition [11]. The current project used a Persian BERT-family checkpoint for text regression.

For structured features, CatBoost was appropriate because it is a gradient-boosted decision-tree method designed to handle categorical variables and reduce prediction shift from target leakage through ordered boosting and ordered categorical statistics [12]. For interpretation, SHAP provides additive feature contributions grounded in Shapley-value principles [13]. Active learning was used because it focuses

annotation effort on informative examples rather than uniformly sampling an overwhelmingly low-stress corpus [14].

3.3 Novelty and scope of comparison

A targeted literature search performed for this report found related English-language pregnancy social-media studies, general Persian mental-health text studies, and Persian Transformer resources, but no directly comparable published study combining a newly constructed Ninisite pregnancy-forum dataset, continuous stress

Transformer fusion. This was a targeted search rather than a formal systematic review; accordingly, the novelty claim is limited should be phrased to "to our knowledge" rather than an absolute first.

Direct performance comparison to previous papers is inappropriate because datasets, languages, targets, class prevalence, unit of analysis, and split procedures differ. The most defensible comparisons are internal: branch-versus-fusion performance on the same locked test and descriptive comparison between the historical pre-active-learning stage and the current leakage-safe stage.

4. End-to-End Project Overview

The complete workflow contains two model-building periods. First, an initial tabular model and initial ParsBERT model were fused to score the unlabeled pool and drive active-learning selection. Human or AI-assisted annotation then expanded the labeled dataset and fed a new training cycle. Second, the final leakage-safe project used canonical labels, grouped roles, five OOF folds, final Member 1 and Member 2 branches, and Member C learned fusion.

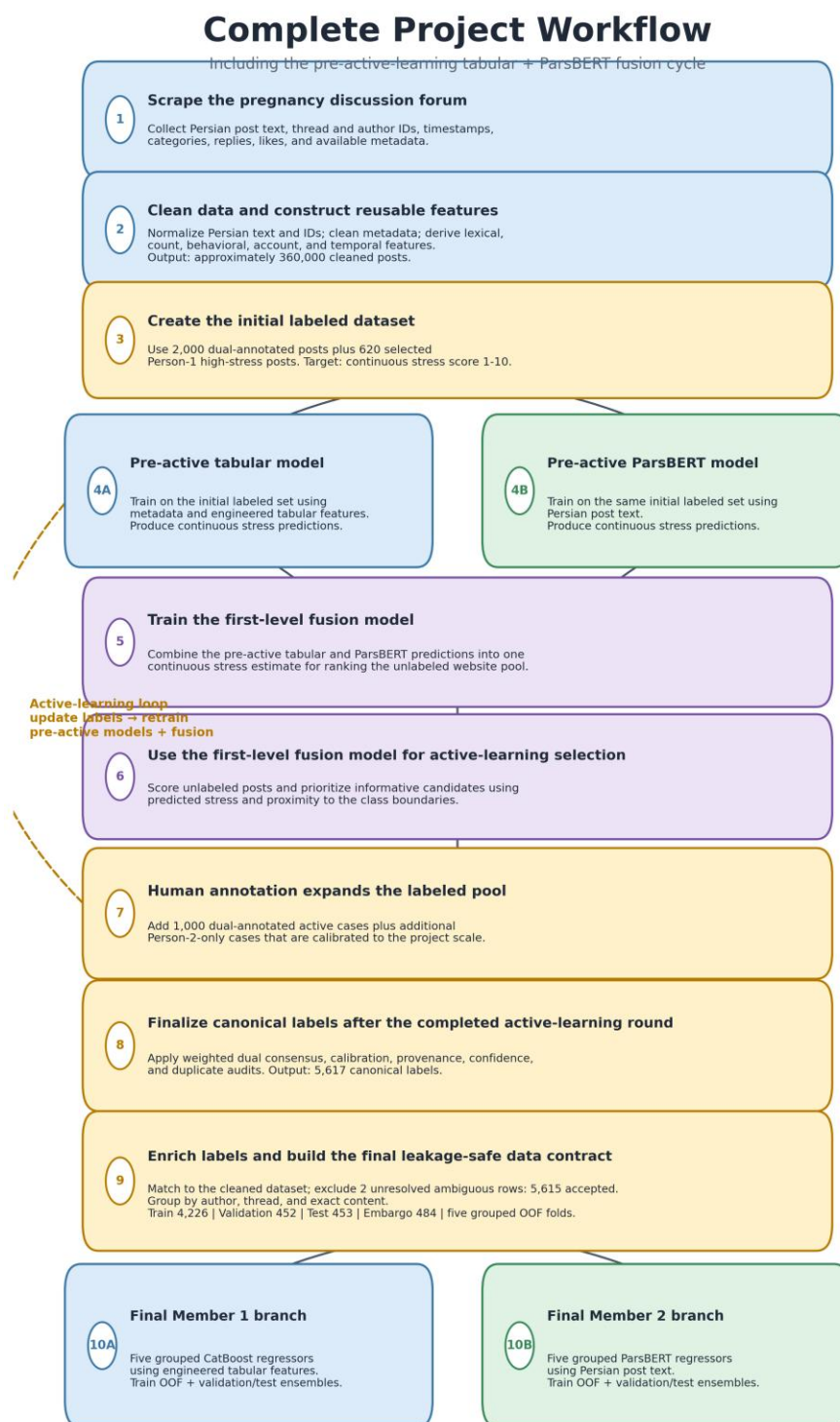
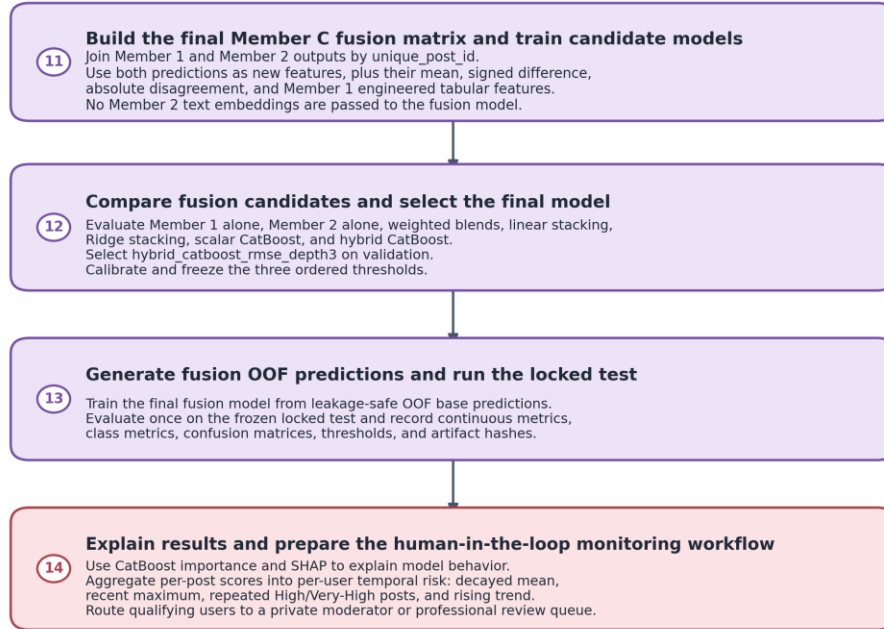


Figure 1. Project workflow, page 1: scraping, cleaning, first-level labels, pre-active tabular and ParsBERT models, first-level fusion, the active-learning loop, canonical reconstruction, and final base-model handoffs.

Source: Project-generated workflow diagram prepared from the team pipeline and reproducibility artifacts.

Complete Project Workflow — Final Modeling and Operation



Future improvement uses new active-learning labels and a fresh untouched confirmation set; the locked test is not reused for tuning.

Figure 2. Project workflow, page 2: final learned fusion, constrained candidate selection, threshold freezing, locked-test evaluation, explainability, temporal monitoring, and human review.

Source: Project-generated workflow diagram prepared from the Member C reproducible pipeline.

5. Scraping, Cleaning, and Exploratory Analysis

5.1 Historical scraping and recovery workflow

Data acquisition was carried out as multiple section/sub-category-specific scraping runs rather than as one monolithic request. Each run began from a known Ninisite forum listing URL, collected thread metadata, paginated through every thread, and then enriched posts with user/profile metadata. Because the site could throttle or return empty pages after many requests, the practical strategy was resumable: save progress, back off, rerun only unfinished or failed work, and keep a shared profile cache. Category and sub-category were normally known from the section configuration at scrape time; three historical source files omitted those two columns and were repaired later during cleaning.

Stage	What was actually implemented	Why it mattered
Section-specific thread discovery	Scrape selected forum/sub-category listing pages separately and collect thread_id, title, URL, and reply_count for each discovered thread.	The source section itself provides category context and allows the acquisition to be resumed file by file.
Thread pagination and post parser	For every thread, paginate all topic pages; anchor on article.topic-post and div.post-message; remove signatures/quoted blocks from post content; extract author/profile, timestamp, user_post_count, likes, reply-to relation, starter status, and derive post_id from share links when available.	Starter posts and replies do not always expose the same identifiers, so the parser keeps enough context for reliable fallback IDs later.

Stage	What was actually implemented	Why it mattered
Checkpoint, retry, and recovery	Completed thread IDs and post rows were checkpointed. Empty/error results were retried with backoff; reruns skipped completed threads and later recovery passes targeted only missing/failed threads.	This made the scrape robust to throttling, timeouts, apparently empty pages, and interrupted long runs.
Robust deduplication during scrape recovery	Use site post_id when it exists; otherwise use the tuple (author, posted_at, content) as a fallback deduplication key.	One representative checkpoint accumulated 1,085,800 rows across repeated runs but reduced to 51,364 robustly deduplicated posts, showing why checkpoint consolidation had to be content-aware.
Profile enrichment and shared cache	Collect unique profile URLs, fetch profile metadata with retries, checkpoint the result, and reuse a shared cache across forum-section runs.	In the representative 2,727-thread run, 15,184 unique profile URLs were observed and the shared cache contained 50,660 profiles; the final rerun had no remaining profile URLs to fetch.
Consolidation into the raw project corpus	The separately scraped forum-section CSVs were combined into combined_all.csv for cleaning and downstream feature construction.	The final combined raw corpus used by the historical cleaner contained 414,994 rows. Three section files were later recognized as missing category/sub_category columns and were repaired from their known source sections.

5.2 Historical cleaning and metadata consolidation

The finalized historical cleaning stage was intentionally conservative. It first removed only structurally unusable text, then repaired metadata and constructed stable identifiers. A crucial fidelity point is that stored post content was not globally Hazm-normalized: surface counts were computed from the preserved raw text, while normalization was applied only inside routines that required token-level normalization or stemming.

Stage	Historical rule	Observed result / rationale
Structural content filter	Start from 414,994 combined rows. Keep non-null, non-whitespace raw content; then require at least 3 whitespace-separated words using content.str.split(). No Hazm normalization is applied before either filter.	414,994 -> 413,180 non-empty posts -> 362,026 posts with at least 3 raw words.
Missing thread-title filter	Inspect missing thread_title values after the text filter. The nine remaining missing-title rows all belonged to thread_id 16400476 and were removed.	362,026 -> 362,017 final cleaned rows.
Stable unique_post_id and duplicate safety	Use the numeric site post_id when present (canonicalized to an integer string so values such as 395597169.0 become 395597169); otherwise use thread_<thread_id>_starter for starters or thread_<thread_id>_<author>_<posted_at> for rare replies without an ID.	This preserves join stability. Repeated site IDs were audited rather than blindly dropped because some historical duplicate IDs corresponded to content-different rows.
Raw-content preservation / local normalization	Keep content as the original scraped string after structural filtering. Apply Hazm locally for education normalization and for lexical tokenization/stemming; the later Transformer branch also normalizes its own text copy before tokenization.	Punctuation, character, question/exclamation, and emoji surface features are therefore measured on the original stored text rather than on a globally rewritten version.

Stage	Historical rule	Observed result / rationale
Category and sub-category recovery	For the three source CSVs that omitted category fields, build the exact key thread_id + author + posted_at and assign the known source-file category/sub-category only to rows that are still missing them.	Historical updates: 21,001 + 37,517 + 40,924 = 99,442 cleaned rows; missing category and sub_category became zero.
Gender recovery	Use scraped profile gender when available; correct the historical marker parser (female iconwoman; male iconuser-01 and older iconman), recover profile URLs from the combined raw file for unresolved cleaned rows, selectively re-scrape unresolved profiles, then apply the historical final assumption that the remaining unresolved profiles were female after manual recheck.	The historical scrape recorded 6,932 checked profiles including 1,437 male profiles. In the final 362,017-row cleaned table, 6,535 post rows are male and 355,482 are female; 20,300 of the female rows come from the documented remaining-unresolved imputation.
Age and education	Parse age from Persian expressions such as a number followed by سال or a pure number; use -1 for unavailable/invalid age. Normalize and map free-text education with Hazm plus regex/keyword rules.	These fields become structured model inputs later; the report treats unknown/cleaned categories as data-engineering choices rather than ground truth.
Signature features and URL cleanup	Compute signature character, punctuation, question, exclamation, emoji, word, and seed positive/negative counts. If a signature is empty, the historical feature values are set to -1. Remove thread_url/profile_url from the modeling table after enrichment; the reproducible cleaning notebook retains profile_url only long enough to replay the historical gender recovery and then drops it.	This preserves the historical feature semantics while keeping raw profile/thread URLs out of the final modeling table.

Historical cleaning checkpoint

414,994 raw combined rows -> 413,180 non-empty raw-content rows -> 362,026 rows with >=3 raw whitespace-separated words -> 362,017 final cleaned posts.

Final metadata checks: missing category = 0; missing sub_category = 0; gender unresolved = 0 under the historical imputation rule; unique_post_id guaranteed for all 362,017 rows.

5.3 Exploratory patterns

Exploratory analysis was performed only after the 362,017-row cleaning contract was established. It was used to understand the forum and design sampling; it was not treated as causal evidence. Longer starter posts tended to have higher average stress-proxy values, but uncertainty increased sharply in the sparse long-post bins. Comment activity also varied strongly by hour and sub-category. These patterns motivated post-length, temporal, category, and interaction features in the structured branch, while the proxy itself remained a sampling aid rather than the final 1-10 supervised target.

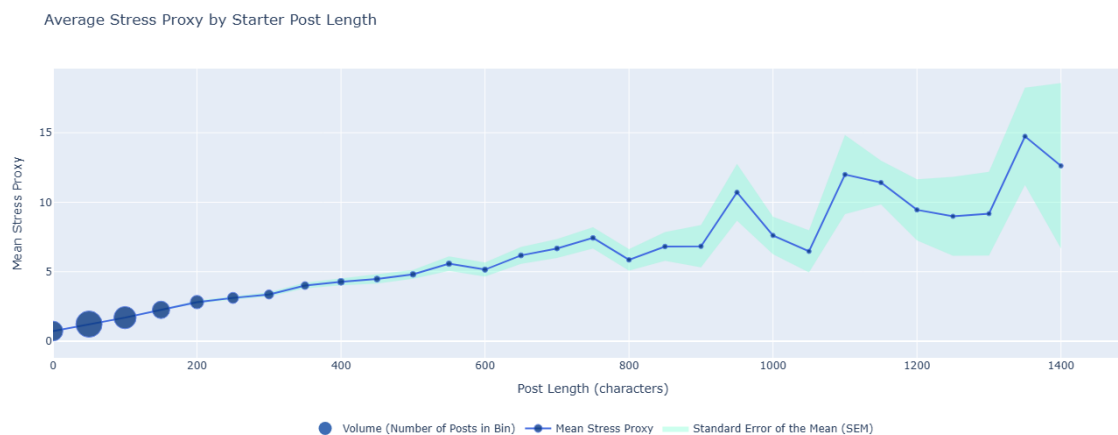


Figure 3. Average exploratory stress proxy by starter-post length. Marker size represents the number of posts in a bin and the

Source: Project-generated EDA figure; the proxy is an exploratory sampling signal, not the final 1-10 target.

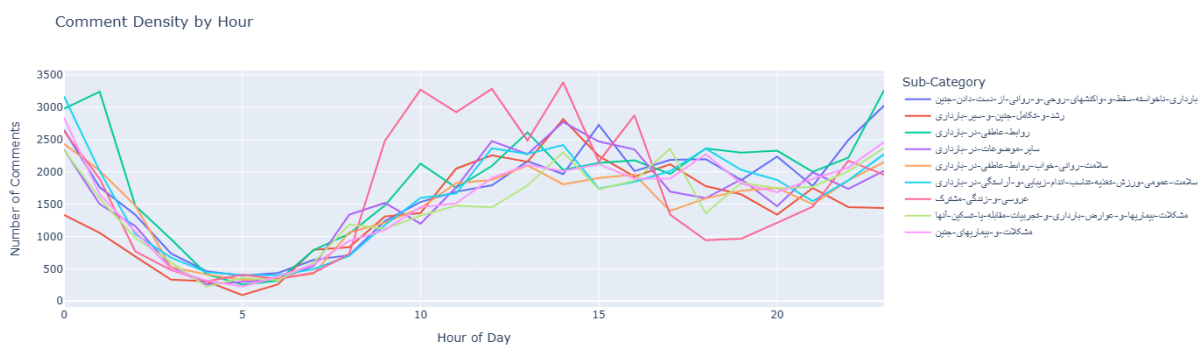


Figure 4. Comment density by hour of day for selected sub-categories. The curves show substantial temporal and category

Source: Project-generated from cleaned forum metadata.

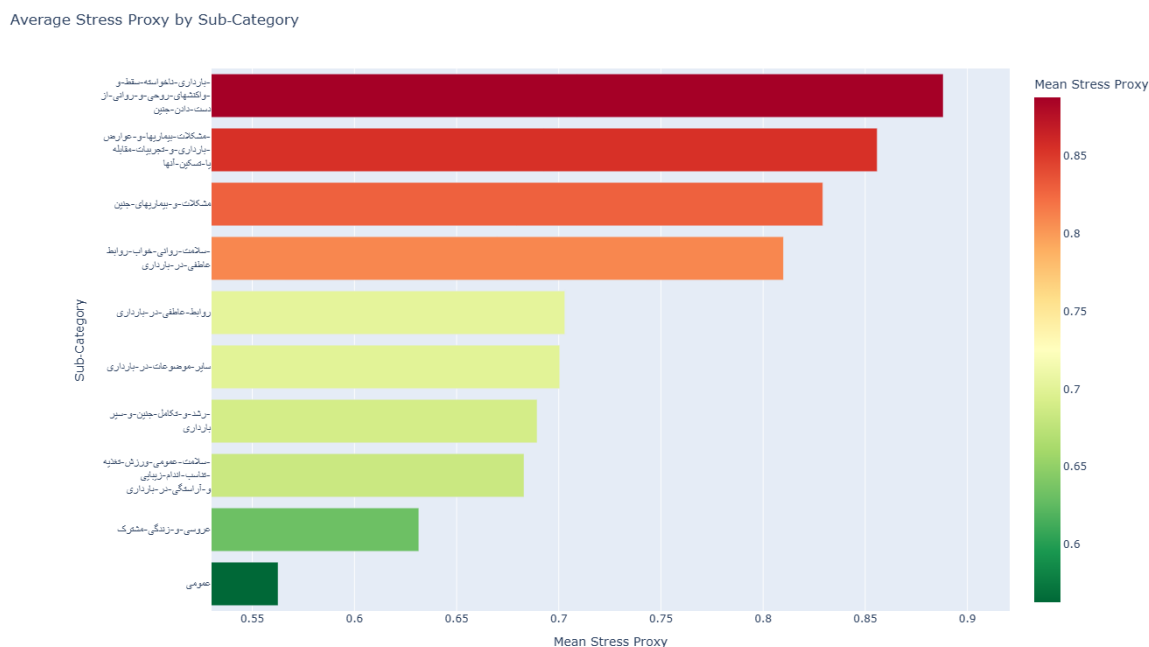


Figure 5. Average exploratory stress proxy by sub-category. The ranking illustrates domain heterogeneity and supported

Source: Project-generated EDA figure; category averages are descriptive and are not interpreted as individual risk.

6. Stress Proxy, Annotation, and Active Learning

6.1 Differential positive/negative lexicon construction

The proxy stage began only after cleaning, because the project initially had a large Persian corpus but essentially no stress labels. Small hand-written positive and negative seed lexicons supplied the first weak signal. Seed counts were computed with local Persian normalization/tokenization, while punctuation and emoji counts came from the preserved raw content. The initial proxy was used only to rank and partition unlabeled posts well enough to induce a broader vocabulary; it was never treated as ground truth and was never passed to the final stress models.

Initial weak stress proxy

$\text{stress_proxy_initial} = 4 * \text{post_neg_count_temp} + 2 * \text{post_neg_emoji} + 0.5 * \text{post_excl_count} + 1.2 * I(\text{post_question_count} > 0) - 0.2 * \text{post_pos_count_temp} - 0.5 * \text{post_pos_emoji}$

The indicator $I(.)$ contributes once when a post contains at least one question mark; starter/reply status is audited separately and is not part of the proxy.

Differential lexicon rule

Negative candidate: $\text{high_count} \geq 10$ and $[\text{low_count} = 0 \text{ or } \text{high_count} / \text{low_count} > 2]$.

Positive candidate: $\text{low_count} \geq 10$ and $[\text{high_count} = 0 \text{ or } \text{low_count} / \text{high_count} > 2]$.

After manual noise filtering, retained candidates were stemmed again exactly as in the legacy notebook before final counts were recomputed.

Using the initial proxy, the upper-quartile high reference contained 96,038 posts (initial threshold 1.2) and the low reference was the 162,000 lowest-proxy posts. Candidate stems required at least 10 occurrences on their associated side and a >2 frequency ratio (or zero count on the opposite side), followed by manual removal of obvious neutral/noisy terms. The historical run produced 584 raw negative and 2,099 raw positive candidates, then 565 negative and 2,081 positive stems after filtering. The retained candidates were stemmed again exactly as in the legacy notebook before the broader post_neg_count and post_pos_count features were recomputed.

Refined stress proxy

$\text{stress_proxy} = 4.0 * \text{post_neg_count_temp} + 0.5 * \text{post_neg_count} + 2.0 * \text{post_neg_emoji} + 1.5 * \text{clip}(\text{post_excl_count}, 0, 6) + 0.5 * \text{clip}(\text{post_question_count}, 0, 5) - 0.3 * \text{post_pos_count_temp} - 0.08 * \text{post_pos_count} - 0.5 * \text{post_pos_emoji}$

Seed negative evidence keeps the strongest weight; broader induced vocabulary receives a smaller weight; punctuation is clipped so repeated marks cannot dominate the score; positive language offsets the score.

On the historical 362,017-post reference run, the refined proxy had mean 1.026, standard deviation 2.708, median 0.18, 75th percentile 0.94, 95th percentile 5.50, minimum -47, and maximum 86. Negative scores are possible because positive lexical/emoji evidence subtracts from the heuristic. The distribution is therefore highly concentrated near low scores with a sparse long right tail, which is exactly why random annotation would have been inefficient.

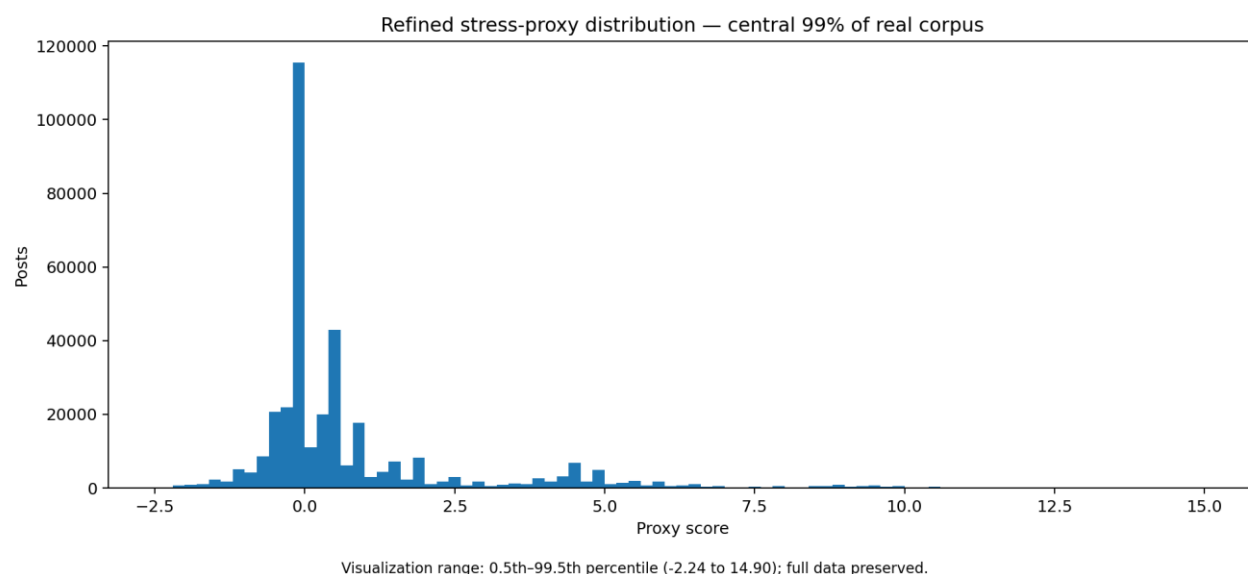


Figure 6A. Refined stress-proxy distribution for the central 99% of the real 362,017-post cleaned corpus. The display trims only the plotting range (0.5th-99.5th percentiles); the full data remain preserved.

Source: Project-generated from the finalized historical-faithful stress-proxy reconstruction. The reconstruction reproduces the algorithm and key checkpoints; minor aggregate drift can remain because the legacy Hazm version and Python-set emoji-regex ordering were not fully pinned.

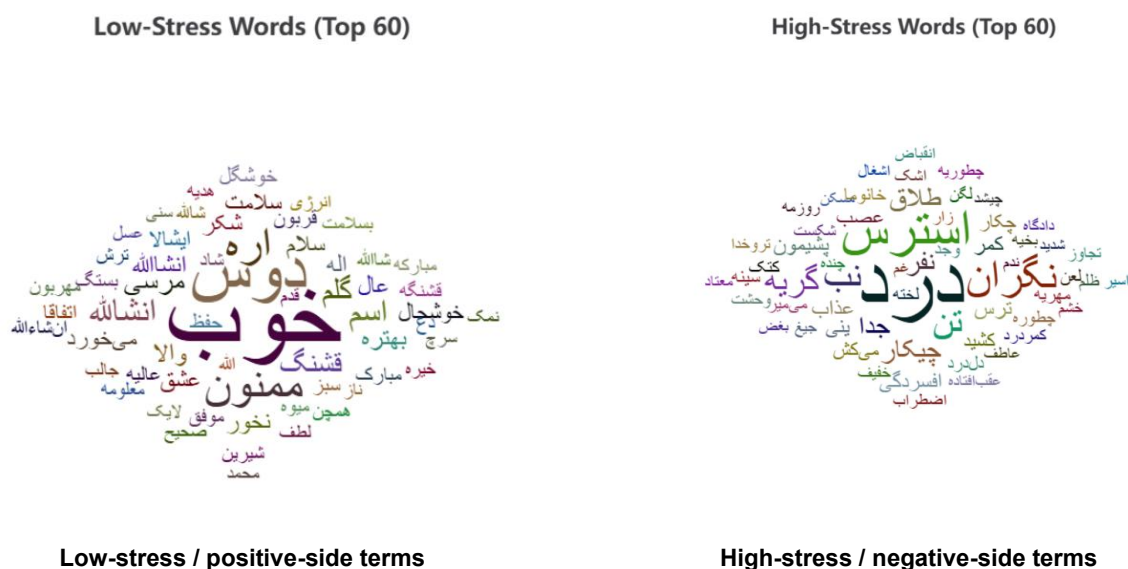


Figure 6B. Word clouds from the differential lexicon analysis used in the first presentation.

Source: Project-generated differential-lexicon word clouds. Font size reflects differential prominence, not causal or clinical weight.

6.2 Sampling for the initial annotation set

The historical notebooks contain several sampling experiments, so the retained annotation files rather than any regenerated sample are the provenance authority. The intended final proxy-guided procedure used three bands: a low band approximately from -0.5 to 0.2, a middle band above 0.2 and below the high threshold, and a high band at or above the refined-proxy 95th percentile (about 5.5 in the historical reference). The documented target sample sizes were 500 low, 600 middle and 900 high posts (2,000 total). This design enriched both the high-stress tail and the ambiguous middle region instead of sampling only extremes. The finalized reproducibility notebook can regenerate this intended procedure, but its newly generated templates are reconstruction outputs and should not replace the retained historical annotation files.

Evaluation-scope caveat

Because sampling deliberately enriched stressful posts, the labeled dataset and final test set must not be used to estimate the natural prevalence of stress on Ninisite.

6.3 AI-assisted multi-dimensional annotation

Two independent commercial large-language-model assistants—OpenAI ChatGPT, accessed through a ChatGPT Plus account, and Anthropic Claude—were used as AI-assisted annotators [20,21]. Each assistant independently scored stress, anxiety, and depression using the project's predefined annotation instructions. Stress was the only target used by the final modeling project. Anxiety and depression were retained to create a richer research resource and to encourage separation of related constructs rather than treating every depressive or anxious expression as stress. These auxiliary labels were never used as features or targets in the final stress model and should not be introduced into the final pipeline. The exact historical model versions used in the original labeling sessions were not consistently recorded and should not be inferred retrospectively.

Labeling limitation

LLM-generated scores are structured AI-assisted annotations, not clinical ground truth. Future work requires expert review and validation against established perinatal screening instruments.

6.4 Consensus and calibration

The original dual labels and the 1,000 dual-labeled active-learning cases used a weighted consensus in which the trusted annotator received weight 1.7 and the second annotator weight 1.0:

Weighted consensus

$$\text{consensus} = (1.7 \times \text{Annotator 1} + 1.0 \times \text{Annotator 2}) / 2.7$$

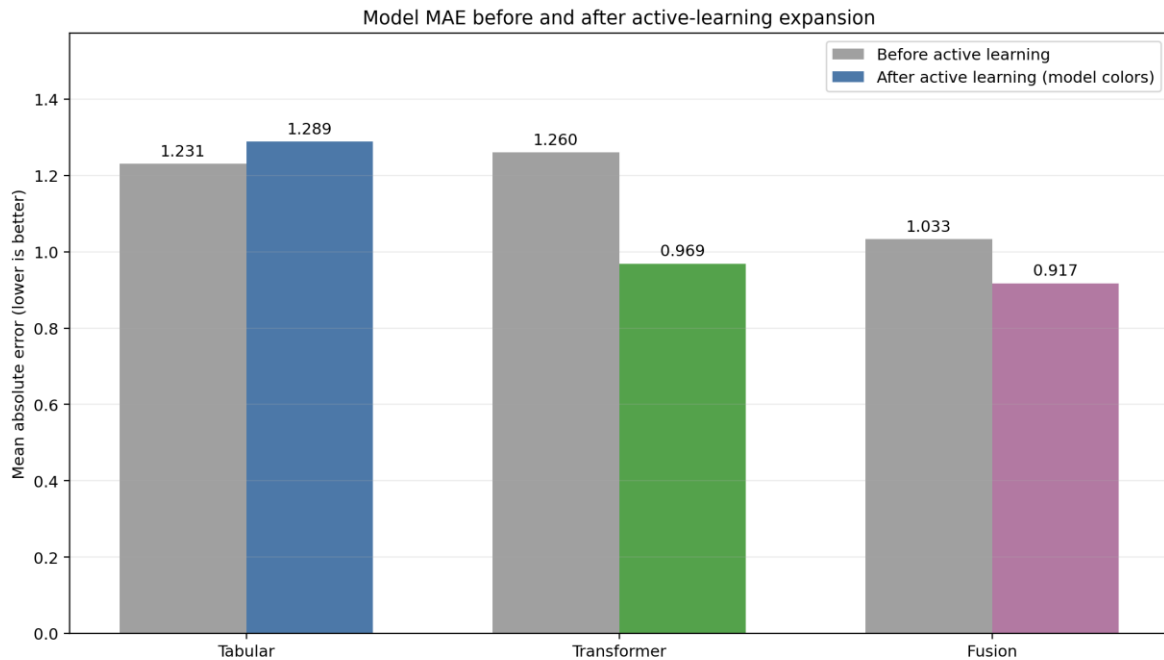
For the remaining active-learning cases with only the second annotation, Person 1 was estimated from Person 2 on the 1,000 overlap rows using ordinary least squares. The reconstructed historical mapping was estimated_Person1 = $0.6459265 \times \text{Person2} + 0.5037576$, followed by the same weighted consensus. This is algebraically equivalent to final = $0.7770648 \times \text{Person2} + 0.3171807$. Five-fold cross-validation on the overlap produced MAE 0.3913 against the observed weighted consensus, slightly better than the tested isotonic/ordinal mapping (MAE 0.3978).

6.5 Canonical labels and active-learning loop

Label source	Canonical rows before enrichment	Role in the label set
Original dual	2,000	Strongest provenance: both annotators available
Original Person-1 high-selected	622	Recovered high-stress cases from the historical pipeline
Active dual	1,000	Both annotators available in the active-learning round
Active Person-2 calibrated	1,995	Single-rater active labels calibrated through the overlap
Total unique canonical posts	5,617	After resolving five IDs repeated across rounds by provenance

The active-learning cycle was not a single isolated sampling step. The initial tabular and ParsBERT models produced predictions, a first-level fusion ranked the unlabeled pool, newly selected posts were annotated, and the expanded labels were used for the final modeling cycle. In the final reproducible project, a new

active-learning ranking module again prioritizes posts near boundaries, posts where Member 1 and Member 2 disagree, and high-risk tail cases for a future annotation round.



Historical pre-active-learning and current post-active-learning values use different split sizes, label pools and leakage controls. The comparison shows project-stage change, not a controlled causal estimate of active learning.

Figure 7. Historical pre-active-learning and current post-active-learning MAE by model family. The Transformer and fusion improved, while the tabular-only branch did not.

Source: Member C report figure generated from historical project metrics and the current locked test. The stages use different label pools, split sizes, and leakage controls; this is descriptive, not a causal estimate of active learning.

Model family	Before active learning MAE	After active learning MAE	Relative change
Tabular	1.2306	1.2889	+4.7% (worse)
Transformer	1.2604	0.9689	-23.1%
Fusion	1.0328	0.9174	-11.2%

7. Member 1 - Tabular Modeling

7.1 Role and design rationale

Member 1 developed the structured-data branch of the system. The branch predicted a continuous stress score from post, user, category, temporal, behavioral, signature, and predefined lexical/count signals without modeling raw content or thread titles as text. Before active learning, the preserved Member 1 workflow operated on a smaller historical train/validation/test snapshot and explored boosted-tree tabular models; the later weighted CatBoost version supplied a continuous tabular score to the first fusion stage used to rank unlabeled posts for active-learning selection. After the expanded annotation stage, the final Member 1 branch was rebuilt on the frozen Member C handoff: 4,226 training rows, 452 validation rows, 453 locked-test rows, and 484 embargo rows that were never used for modeling. The final feature engineer reproduced 67 ordered features covering interaction metadata, account history, posting time, forum category, signature statistics, post-length/punctuation/emoji counts, positive/negative lexical counts, and derived ratios. CatBoost was selected because gradient-boosted trees are strong on small-to-medium structured datasets and CatBoost handles mixed numerical and categorical inputs natively while preserving missing numerical values [12]. Training used the supplied training_sample_weight and the exact grouped five-fold assignments. The fusion handoff consisted of one held-out OOF prediction for every training row,

five-fold mean ensemble predictions for validation and test, and the same engineered tabular feature tables aligned by unique_post_id. These outputs are leakage-safe research predictions, not clinical diagnoses.

7.2 Final model configuration

Field	Final Member 1 implementation
Accepted input files	member1_handoff.csv.gz — 5,615 rows / 52 columns; contains all frozen roles: train 4,226, validation 452, test 453, embargo 484. Important columns include unique_post_id, final_stress, clinical_class, model_role, oof_fold, training_sample_weight, metadata and predefined post/signature count features. modeling_manifest_v2.csv — 5,615 rows / 25 columns; authoritative ID, role, group and fold manifest including unique_post_id, group_id, author, thread_id, content_hash, final_stress, clinical_class, model_role and oof_fold. member1_feature_contract.json — contract forbidding raw text, stress_proxy and target/split/weight leakage fields. oof_predictions_template.csv — schema: unique_post_id, fold, true_stress, prediction, prediction_std_optional. holdout_predictions_template.csv — schema: unique_post_id, model_role, true_stress, prediction, prediction_std_optional.
Feature preparation	67 ordered features. Numeric: likes/activity/account counts, age, children, date components, signature/post counts and derived rates. Categorical: gender, category, sub_category, education_clean, posted_weekday_name, posted_daypart, handled natively by CatBoost. Missing numeric values remain NaN. age_num=-1 or ages outside 12–80 become NaN and age_missing marks absence. Missing categorical values become __MISSING__. Signature -1 denotes no signature: values become 0 and has_signature=0. log1p transforms: likes, user_post_count, posts_per_account_year. Ratios include likes_per_100_posts, positive/negative word ratios, punctuation per 100 characters, question/exclamation/emoji per 100 words, and temporary positive/negative ratios. Jalali posted_at/join_date produce date, weekday/daypart, weekend/night indicators, hour/month sine-cosine encodings, account_age_days/years and posting rate. category and education_clean arrive already cleaned upstream and are preserved as categorical strings. Excluded from the model include content, thread_title, stress_proxy, final_stress, clinical_class, role/fold/group/label/weight columns, author/thread IDs, unique_post_id, raw_reply_to (converted only to has_reply), and redundant gender_code.
Model/checkpoint	CatBoostRegressor [12]. Five saved fold models: models/fold_0/model.cbm, models/fold_1/model.cbm, models/fold_2/model.cbm, models/fold_3/model.cbm, models/fold_4/model.cbm. An additional models/final_full_train/model.cbm is saved for completeness, but official OOF/validation/test handoffs and default future inference use the five-fold ensemble.
Hyperparameters	loss_function=RMSE; eval_metric=RMSE; iterations=558; depth=8; learning_rate=0.025; l2_leaf_reg=10; random_strength=0.3; bootstrap resolved by the serialized CPU models as MVS; bagging_temperature=0.5 is present in project configuration but is not an active MVS parameter; early stopping disabled (fixed 558 iterations, use_best_model=False, held-out fold not used for early stopping); random seeds 42,43,44,45,46 for folds 0–4; thread_count=4 in training configuration; task_type=CPU.
Fold procedure	The supplied grouped folds were used exactly; no new random split was created. Held-out sizes were fold 0=844, 1=844, 2=848, 3=847, 4=843. For each fold, CatBoost trained on the other four folds using training_sample_weight and predicted only the held-out fold. The five held-out blocks were concatenated to create complete OOF predictions for all 4,226 training rows. Each validation and test row was predicted by all five fold models and the arithmetic mean was stored as prediction; prediction_std_optional stores the population standard deviation across fold predictions. Group checks confirmed no author, thread, content_hash or group_id crossing between train folds or modeled roles.
Validation metrics	Continuous: MAE=1.2657; RMSE=1.5924; Pearson r=0.6886; Spearman rho=0.6577. Frozen prediction thresholds: 2.9291, 4.2652, 6.2155 (optimized before test inspection). Accuracy=0.6195; macro F1=0.5284; Very-High PR-AUC=0.5560. Recall: Low 0.6902, Moderate 0.5373, High 0.4444, Very High 0.4400.
Locked-test metrics	Continuous: MAE=1.2889; RMSE=1.6556; Pearson r=0.6727; Spearman rho=0.6261. Accuracy=0.6004; macro F1=0.4815; Very-High PR-AUC=0.4292. Per-class P/R/F1: Low 0.8750/0.7071/0.7821; Moderate 0.1695/0.2985/0.2162; High 0.4225/0.4762/0.4478; Very High 0.5000/0.4615/0.4800.
Artifacts delivered to Member C	data/handoff/member1_oof_predictions.csv (4,226 rows; unique_post_id, fold, true_stress, prediction, prediction_std_optional); data/handoff/member1_validation_predictions.csv (452 rows; unique_post_id, model_role, true_stress, prediction, prediction_std_optional); data/handoff/member1_test_predictions.csv (453 rows; same holdout schema); data/handoff/member1_train_engineered_features.csv.gz, member1_validation_engineered_features.csv.gz, member1_test_engineered_features.csv.gz (67 engineered features in exact order); data/handoff/member1_handoff_manifest.json; data/processed/member1_exact_feature_order.csv; data/processed/member1_preprocessor.joblib; five .cbm fold models and the Member1_Final_Reproducible_Project package. Member C receives the continuous prediction column named prediction plus engineered tabular features aligned by unique_post_id.
Limitations	The branch depends on structured metadata and predefined count/signal columns and therefore cannot

Field	Final Member 1 implementation
	fully represent semantic meaning, negation or context. User metadata can be missing or inaccurate, and historical forum/annotation patterns may introduce demographic or temporal bias. The original upstream lexicon/extractor resources for reconstructing positive/negative count columns from raw text were not included in the accepted handoff, so exact raw-post-only inference requires that upstream service. The tabular branch is materially weaker than the final Transformer/fusion system on the same locked test (Member 1 MAE 1.2889 versus final fusion MAE 0.9174 in the project report). It also misses several internal class-recall objectives and should not be interpreted as a stand-alone diagnostic model.

7.3 Training, evaluation, explainability, and handoff

7.3.1 Pre-active-learning tabular model

The preserved pre-active-learning Member 1 notebook used the smaller historical train.csv, val.csv and test.csv snapshot (1,561/335/335 rows before two overlapping training IDs were removed). Earlier experiments compared LightGBM and CatBoost and tested target weighting and threshold calibration. The later preserved early-stage configuration used a weighted CatBoost regressor with inverse-frequency weights moderated by $\gamma=0.35$, categorical features handled natively, and continuous stress as the prediction target. Its purpose was not to create a final four-class classifier; it produced a continuous tabular score that could be joined by unique_post_id with the first Persian text-model score. The first-level fusion then ranked unlabeled examples for the active-learning cycle described in Section 6. The early branch already used enriched interaction, temporal, demographic, post-count, signature-count and lexical-count signals, but it predated the final connected-component role assignment and the authoritative grouped OOF folds. Its train/validation/test split was therefore historical evidence only. The final project does not reuse those rows, models or thresholds as its evaluation basis; the post-active-learning package starts again from the frozen Member C handoff.

7.3.2 Final input and feature-engineering pipeline

The final handoff contains 5,615 labeled records, but Member 1 transforms and models only the 4,226 train, 452 validation and 453 test rows; 484 embargo rows remain untouched. A stateless Member1FeatureEngineer validates the required upstream columns and emits exactly 67 features in a fixed order. Raw content and thread_title are never passed to CatBoost as text. The predefined post/signature count columns are accepted as numeric upstream signals, consistent with the feature contract. Raw-content preservation and the original count extraction belong to the upstream data pipeline described earlier in the report; normalization is local to the routines that require it, and Member 1 does not retokenize raw posts during final training [15]. category and education_clean are likewise treated as already-normalized upstream categorical fields and missing values are mapped to `__MISSING__`. Numerical NaN values remain NaN because CatBoost supports missing numerical values natively [12].

Table 7.1. Member 1 feature families.

Feature family	Count	Examples / transformations
Post metadata and interaction	5	likes, log1p_likes, is_starter, has_reply, likes_per_100_posts
User/account	12	user_post_count, age_num + age_missing, children_count, join-date fields, account age, posts/year and log rate
Temporal posting	15	Jalali date parts, hour, weekday/daypart, weekend/night indicators, hour/month sine-cosine
Forum/demographic categorical	4	gender, category, sub_category, education_clean
Signature signals	11	has_signature plus character, punctuation, word, emoji and positive/negative counts
Post lexical/count signals	20	length/punctuation/emoji/positive-negative counts plus eight normalized ratios
Total	67	Exact order stored in data/processed/member1_exact_feature_order.csv

Age handling was deliberately conservative: age_num=-1, missing ages, and values outside 12–80 become NaN, while age_missing preserves the fact that age was absent. For signatures, a row whose signature fields are all -1 is interpreted as having no signature; those values become zero and has_signature=0. Derived post ratios normalize positive/negative word counts and punctuation/emoji counts by post length. Temporal engineering uses the Jalali post and join timestamps to reconstruct weekday, daypart and account age. The exact 67-feature order is an artifact of the project rather than an inferred order, preventing silent column drift during new-post inference.

7.3.3 CatBoost regression model

The final model family is CatBoostRegressor [12], a gradient-boosted decision-tree algorithm well suited to heterogeneous tabular data. The response variable is `final_stress`, modeled continuously on the 1–10 project scale. Six categorical features—`gender`, `category`, `sub_category`, `education_clean`, `posted_weekday_name` and `posted_daypart`—are passed directly to CatBoost rather than one-hot encoded. Training consumes the frozen `training_sample_weight` supplied by Member C; Member 1 does not recompute class or annotation weights. This preserves the project-level weighting policy while keeping validation and test metrics unweighted. Every fold uses 558 trees, depth 8, learning rate 0.025, L2 leaf regularization 10 and random strength 0.3. The serialized CPU models resolve to MVS bootstrap. The configuration contains `bagging_temperature=0.5`, although this parameter is not active for MVS. Training is fixed-iteration with no early stopping and `use_best_model=False`. Seeds are deterministic: 42+fold, giving 42–46 across folds, with four CPU threads configured. All five fold models are saved as native `.cbm` files, and an additional full-train `.cbm` is archived but is not used to generate the official holdout predictions.

7.3.4 OOF and ensemble prediction generation

Out-of-fold (OOF) prediction is essential because a fusion model must not learn from a base-model score produced on the same row that trained that base model. The supplied `modeling_manifest_v2.csv` assigns every training row to exactly one of five connected-component-safe folds. For fold *k*, Member 1 trains CatBoost on the other four folds and predicts only fold *k*. The five held-out prediction blocks are concatenated, producing exactly one OOF prediction for each of the 4,226 training IDs. Validation and locked-test rows are never used to fit any fold model. Instead, each holdout row is scored by all five models; the arithmetic mean becomes prediction and the population standard deviation becomes `prediction_std_optional`. `unique_post_id` is retained as the alignment key throughout, so file ordering cannot alter the fusion join. The input audit verified zero cross-fold or cross-role leakage for `group_id`, `author`, `thread_id` and `content_hash`.

Member 1 leakage-safe tabular workflow

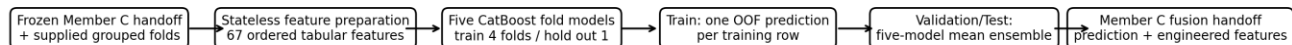


Figure 7.1. Leakage-safe Member 1 training and handoff workflow.

Training rows receive one held-out prediction, while validation and test predictions are averages across all five fold models. This separation is the key leakage control required by the downstream fusion model.

Source: project-generated from Member 1 model outputs.

7.3.5 Member 1 results

Continuous performance is summarized below using the saved Member 1 metrics artifact. OOF metrics estimate generalization within the training population under the supplied grouped folds; validation was used for model/threshold decisions; the locked test was not used to tune the CatBoost model or prediction thresholds. The final tabular branch is useful as a complementary structured signal, but its locked-test error is higher than the final learned fusion reported elsewhere in this document. Metric calculation and reporting used NumPy [22], pandas [17], SciPy [23], scikit-learn [16], and Matplotlib [24] as documented in the reproducible package.

Table 7.2. Continuous Member 1 performance.

Dataset	MAE	RMSE	Pearson r	Spearman rho
Training OOF	1.2663	1.5782	0.6943	0.6914
Validation	1.2657	1.5924	0.6886	0.6577

Dataset	MAE	RMSE	Pearson r	Spearman rho
Locked test	1.2889	1.6556	0.6727	0.6261

Table 7.3. Validation class metrics at frozen prediction thresholds.

Class	Precision	Recall	F1	Support
Low	0.8650	0.6902	0.7678	297
Moderate	0.2647	0.5373	0.3547	67
High	0.4375	0.4444	0.4409	63
Very High	0.7333	0.4400	0.5500	25

Table 7.4. Locked-test class metrics at the same frozen thresholds.

Class	Precision	Recall	F1	Support
Low	0.8750	0.7071	0.7821	297
Moderate	0.1695	0.2985	0.2162	67
High	0.4225	0.4762	0.4478	63
Very High	0.5000	0.4615	0.4800	26

Validation accuracy=0.6195, macro F1=0.5284, Very-High PR-AUC=0.5560. Locked-test accuracy=0.6004, macro F1=0.4815, Very-High PR-AUC=0.4292. The operational true classes remain defined by true-score cutoffs 3, 5 and 7, while the frozen prediction cutoffs are 2.9291, 4.2652 and 6.2155. The locked-test results show that the tabular branch alone does not satisfy all four project recall floors; its role is therefore complementary to the stronger text/fusion system rather than stand-alone decision making.

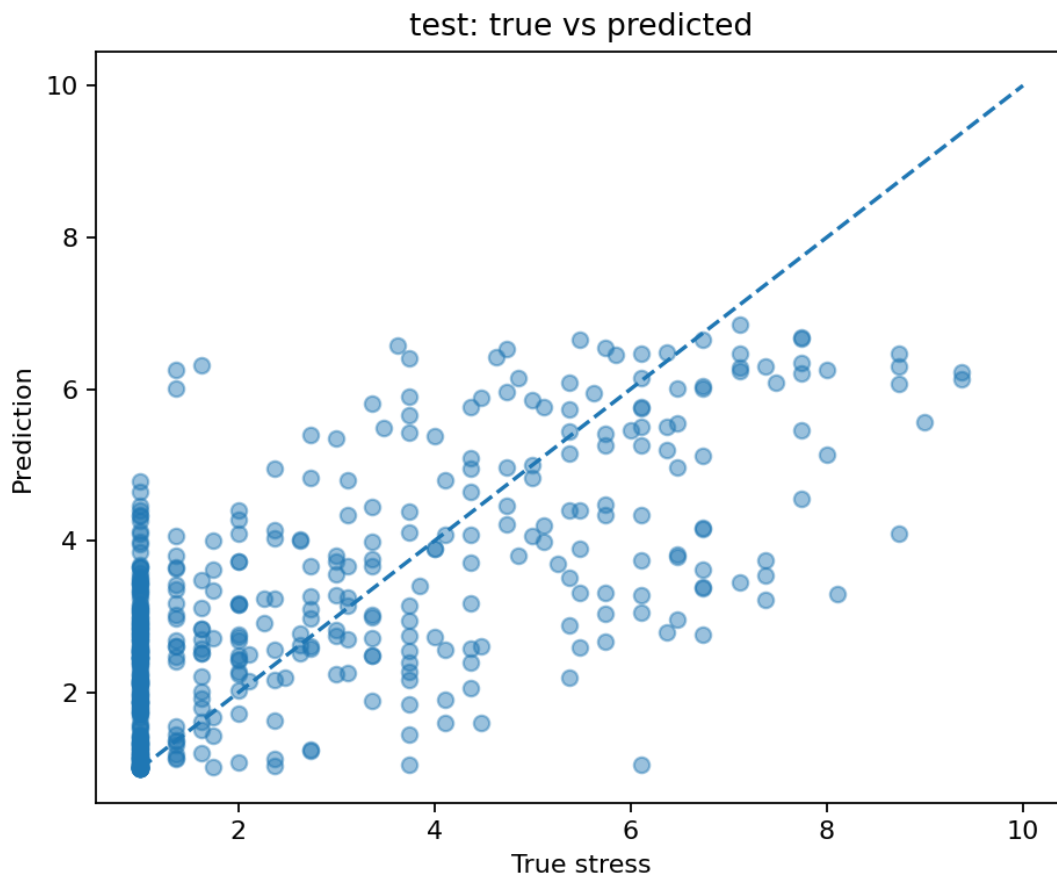


Figure 7.2. Locked-test true versus predicted continuous stress for Member 1.

Predictions follow the overall stress ordering but compress the most extreme scores toward the center, which contributes to the weaker High and Very-High class recall compared with the final fusion.

Source: project-generated from Member 1 model outputs.

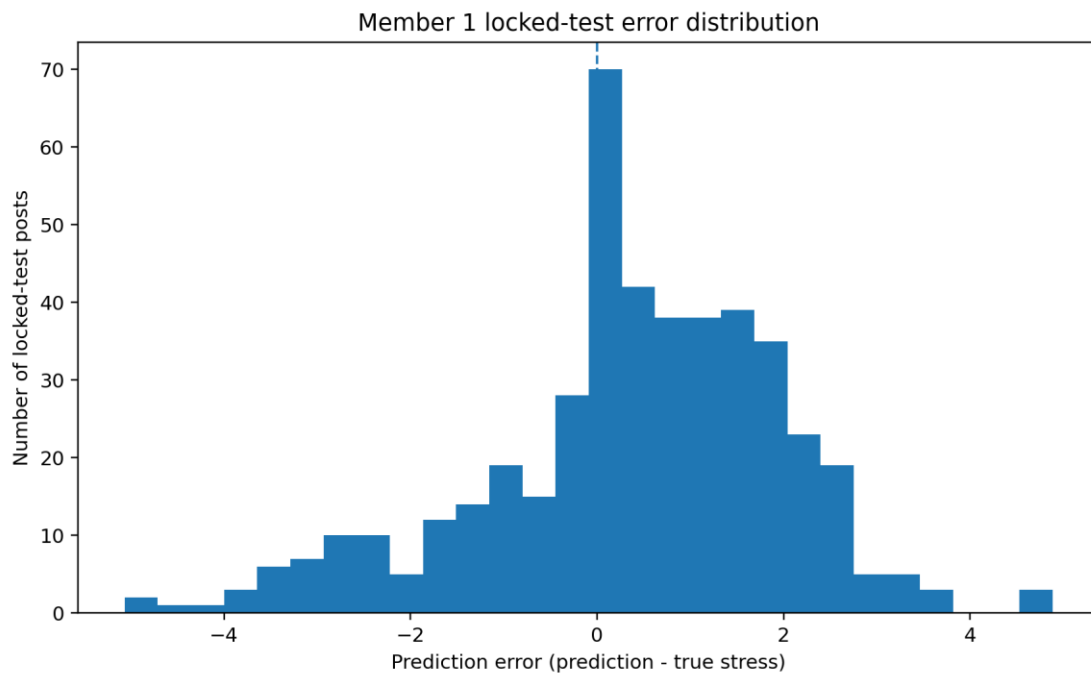


Figure 7.3. Distribution of signed Member 1 prediction error on the locked test.

Most errors lie near zero, but the tails show that a subset of posts are substantially under- or over-predicted; this motivates the downstream fusion and human-review framing.

Source: project-generated from Member 1 model outputs.

7.3.6 Explainability

Global CatBoost feature importance was calculated independently for the five fold models and averaged after training. It was used for interpretation and reporting only, not for formal feature selection. The largest mean importances were `is_starter` (16.59), `post_neg_emoji` (7.93), `post_char_count` (7.26), `post_neg_count` (7.26), `post_pos_word_ratio` (4.94), `post_temp_neg_ratio` (3.74), `post_word_count` (3.71) and `sub_category` (3.26). These results indicate that the structured branch combines behavioral context with post-scale and positive/negative signal counts rather than relying on one feature family. CatBoost importance measures model usage, not causal influence. Native CatBoost SHAP code is implemented in `src/explain_model.py`, but the executed final reproduction recorded `shap_run=false`; therefore no Member 1 SHAP figure is presented here and SHAP was not used for Member 1 feature selection.

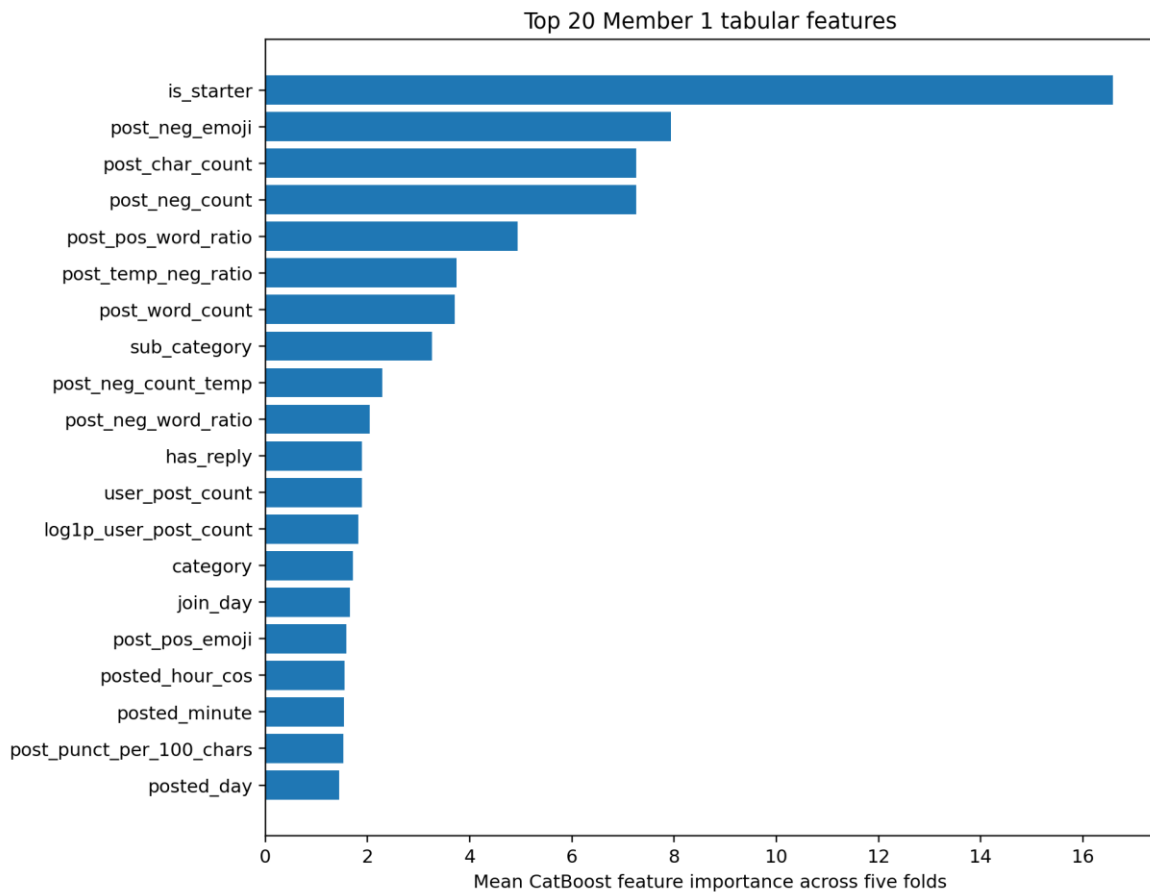


Figure 7.4. Top 20 mean CatBoost feature importances across the five Member 1 fold models.

Starter/reply context and negative-language count signals are prominent, while post length, category and account-activity variables add complementary structured information. Importance is descriptive of the fitted model and should not be interpreted causally.

Source: project-generated from Member 1 model outputs.

7.3.7 Handoff to Member C and new-post inference

Member C receives two types of information from Member 1: the continuous tabular prediction and the engineered tabular features themselves. Training fusion rows use `data/handoff/member1_oof_predictions.csv`, whose prediction field is genuinely held out. Validation and test use `member1_validation_predictions.csv` and `member1_test_predictions.csv`, where prediction is the mean of the five fold scores and `prediction_std_optional` records fold disagreement. The engineered tables `member1_train_engineered_features.csv.gz`, `member1_validation_engineered_features.csv.gz` and `member1_test_engineered_features.csv.gz` provide the same 67 features used by CatBoost. All files align through `unique_post_id` and predictions are clipped to `[1,10]`. The five native fold models are included in the reproducible project because future inference uses their ensemble rather than the optional full-train model. For a completely new post, the stateless feature engineer validates the required metadata/count fields, recreates the 67-feature vector in the stored order, each fold model produces one score, and the mean becomes the Member 1 prediction. Exact raw-post-only operation is conditional on the upstream service supplying the predefined post/signature positive-negative count columns, because the original lexicon/extractor resource files were not included in the accepted Member 1 handoff.

7.3.8 Limitations

The tabular branch cannot recover full semantic meaning, sarcasm, negation or long-range linguistic context; two posts with similar counts can express very different experiences. Some user/account metadata are missing or self-reported, so inaccurate age, education or profile fields can add noise. Category and

behavior distributions reflect one forum and one historical period and may shift over time. The positive/negative signal columns are dependent on an upstream extractor whose original lexicon resources are not bundled with the final Member 1 package. Although grouped roles and OOF prediction remove the main author/thread/duplicate-content leakage channels, remaining dataset and annotation biases can still affect generalization. The final locked-test MAE of 1.2889 and class recalls show that the structured branch is weaker than the final Transformer/fusion system, especially near class boundaries. Future improvements could include richer leakage-safe interaction histories, verified upstream extractor versioning, robustness checks for missing metadata, temporal drift evaluation, and new untouched labels rather than further tuning on the opened locked test.

8. Member 2 - Persian Transformer Modeling

8.1 Role and design rationale

Member 2 provides the project's text-only branch. The model predicts the continuous stress score from raw Persian post content while deliberately excluding metadata, engineered count features, stress_proxy, anxiety and depression. This modality isolation makes the branch scientifically interpretable: Member 1 captures structured/contextual signals, while Member 2 captures contextual semantics from language itself. The final Member C stacker can therefore test whether combining two genuinely different information sources improves generalization. Regression is used rather than direct four-class classification because the project target is continuous; the ordered Low/Moderate/High/Very High bins are applied later for evaluation and operational thresholding.

The base model is HooshvareLab/bert-fa-zwnj-base (ParsBERT v3), a Persian-specific BERT-family checkpoint [10,11]. Pretraining supplies a strong Persian language representation before the relatively small stress dataset is seen. Fine-tuning adapts that representation to stress regression. The model is ZWNJ/half-space aware and is paired with Hazm normalization [15] to reduce superficial Persian orthographic variation before tokenization.

8.2 Pre-active-learning text branch

Before the active-learning round, Member 2 trained an exploratory ParsBERT regression model on the historical 1,561/335/335 train/validation/test snapshot. Only content, stress and unique_post_id were used. Text was normalized with Hazm, tokenized with the ParsBERT tokenizer at max_length=256, and fine-tuned with AdamW. The historical model used a weighted MSE objective with a 1.75x under-prediction penalty and early stopping on validation MAE. Its historical test metrics were MAE 1.260, RMSE 1.527 and Pearson r 0.746. The resulting continuous text predictions were combined with the early tabular branch in a first-level fusion, and that initial fusion ranked unlabeled candidates for the active-learning cycle described in Section 6. These early metrics are historical context only and are not reused as the final leakage-safe evaluation.

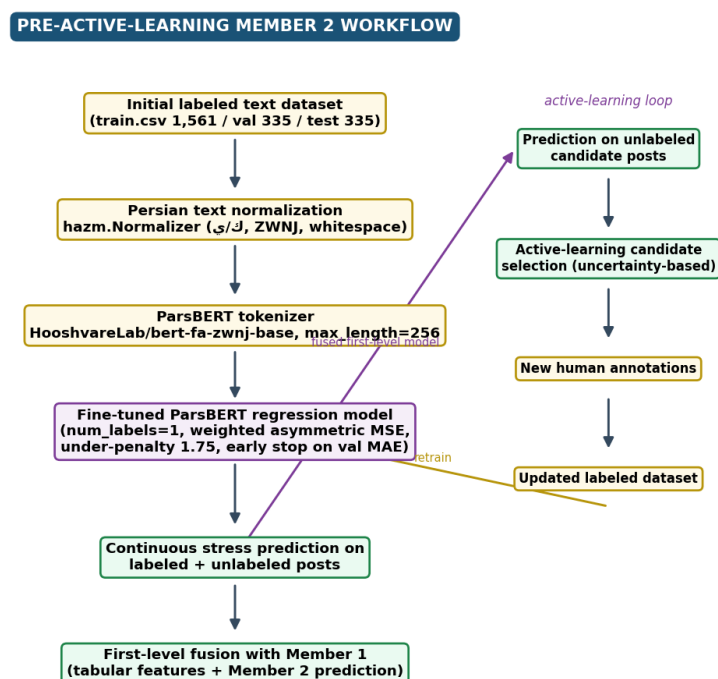


Figure 8.1. Member 2 pre-active-learning workflow and connection to the first fusion/active-learning loop.

Source: Project-generated Member 2 pre-active-learning workflow figure.

8.3 Final data contract and Persian preprocessing

After active learning and canonical label reconstruction, the project froze one leakage-safe handoff: 4,226 training rows, 452 validation rows, 453 locked-test rows and 484 embargo rows that were never modeled. The final Member 2 input contract uses content as the only predictive input, final_stress as the regression target, training_sample_weight only inside the loss, unique_post_id as the join key, clinical_class only for evaluation, and the supplied oof_fold assignments for grouped training. The modeling manifest additionally carries group_id, author, thread_id and content_hash so the notebook can verify that author/thread/duplicate-content groups do not cross frozen folds or roles.

Member 2 applies hazm.Normalizer locally to text and keeps URLs, emoji and punctuation rather than stripping them aggressively. The normalized text is tokenized by the ParsBERT WordPiece tokenizer with truncation and padding to max_length=384. This value was selected from measured token lengths: the 99th percentile is about 262 tokens and only a small number of posts exceed 384, so the setting retains almost all context without paying the memory cost of the rare >1,000-token tail.

Table 8.1. Final Member 2 model and data configuration.

Field	Final Member 2 implementation
Accepted data	member2_handoff.csv + modeling_manifest_v2.csv; train 4,226 / validation 452 / test 453 / embargo 484; five supplied grouped folds (844/844/848/847/843).
Text input	content only; Hazm normalization; ParsBERT tokenizer; max_length=384; truncation=True; padding="max_length"; no stopword removal, stemming, or punctuation/emoji stripping.
Model	HooshvareLab/bert-fa-zwnj-base (ParsBERT v3) -> BertForSequenceClassification(num_labels=1, problem_type="regression"), one continuous output.
Training objective	weighted_asymmetric_mse = training_sample_weight x direction_penalty x error^2; direction_penalty = 1.75 when prediction < target. Project risk-sensitive preference, not a clinical constant.
Optimization	AdamW; learning rate 2e-5; weight decay 0.01; batch 16; linear warmup 10% then linear decay; gradient clipping 1.0; up to 5 epochs per fold

	with patience=2 early stopping; seed 42; CUDA/A10G for full training.
Deterministic delivery	model.eval(); no MC-dropout in the primary handoff. Train uses one held-out OOF prediction per row; validation/test use the mean of all five fold-model predictions.
Primary Member C handoff	Scalar prediction keyed by unique_post_id. CLS 768-D embeddings were exploratory artifacts and are not required by the frozen final fusion.

8.4 Five-fold ParsBERT training and leakage-safe predictions

Each final fold model is trained on four of the five supplied training folds and validated on its held-out fold. The network has one regression output. For training row i with target y_i , prediction $\hat{y}_i$ and supplied weight w_i , the loss is $w_i \times d_i \times (\hat{y}_i - y_i)^2$, where $d_i=1.75$ for under-prediction and 1 otherwise. The sample weight addresses imbalance and annotation provenance, while the directional factor encodes the project preference to penalize false reassurance more strongly. Squared error still makes large mistakes more expensive than small ones. The 1.75 factor is an engineering choice requiring external validation; it is not a medical standard.

The five-fold protocol is essential for stacking. For fold k , the ParsBERT model trained on the other four folds predicts only fold k ; concatenating these held-out blocks gives exactly one out-of-fold prediction for every training row. Member C therefore never sees an in-sample Member 2 training prediction. Validation and test are predicted by all five fold models in eval mode and the arithmetic mean is used as the deterministic text score. Between-model standard deviation is saved separately as a diagnostic, but the frozen primary fusion uses the scalar prediction. The test set is evaluated only after model settings are frozen.

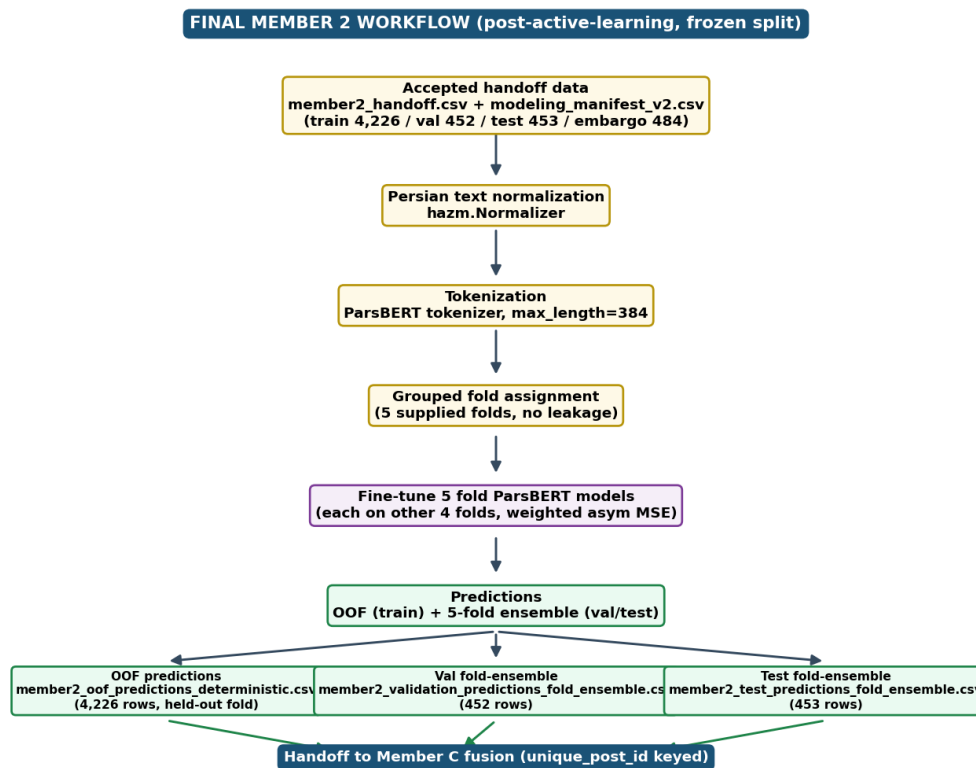


Figure 8.2. Final Member 2 leakage-safe five-fold workflow after active learning.

Source: Project-generated Member 2 final leakage-safe workflow figure.

8.5 Member 2 results

The final deterministic fold ensemble is the strongest individual branch in the project. OOF performance estimates generalization within the training population under the grouped five-fold protocol; validation was available during development; the 453-post locked test is the same final test later used for the fusion

comparison. The exact values below are taken from the executed Member 2 metrics artifacts and the supplied prediction files/confusion matrices: test MAE 0.9689, RMSE 1.3182, Pearson r 0.8251, and Very-High recall 17/26 = 65.38%.

Table 8.2. Continuous Member 2 performance from the final deterministic handoff.

Dataset	N	MAE	RMSE	Pearson r	Spearman rho	Accuracy (4 bins)
Training OOF	4,226	0.9845	1.3193	0.8130	0.8003	0.6488
Validation	452	0.8246	1.1462	0.8596	0.8061	0.7412
Locked test	453	0.9689	1.3182	0.8251	0.7891	0.7285

Table 8.3. Member 2 locked-test four-class metrics using the authoritative true-score bins.

Class	Precision	Recall	F1	Support
Low	0.9179	0.8283	0.8708	297
Moderate	0.3333	0.4478	0.3822	67
High	0.5286	0.5873	0.5564	63
Very High	0.6800	0.6538	0.6667	26

At the project-defined internal recall floors, the text-only branch independently passes Low (82.83% $\geq$ 75%) and High (58.73% $\geq$ 50%), while Moderate (44.78% $<$ 50%) and Very High (65.38% $<$ 75%) remain below their targets. This does not invalidate Member 2: it is a base model rather than the final operating system. The final learned fusion reduces locked-test MAE from 0.9689 to 0.9174 and raises Very-High recall from 65.38% to 76.92%, showing why the structured branch remains useful even though the Transformer carries the strongest single predictive signal.

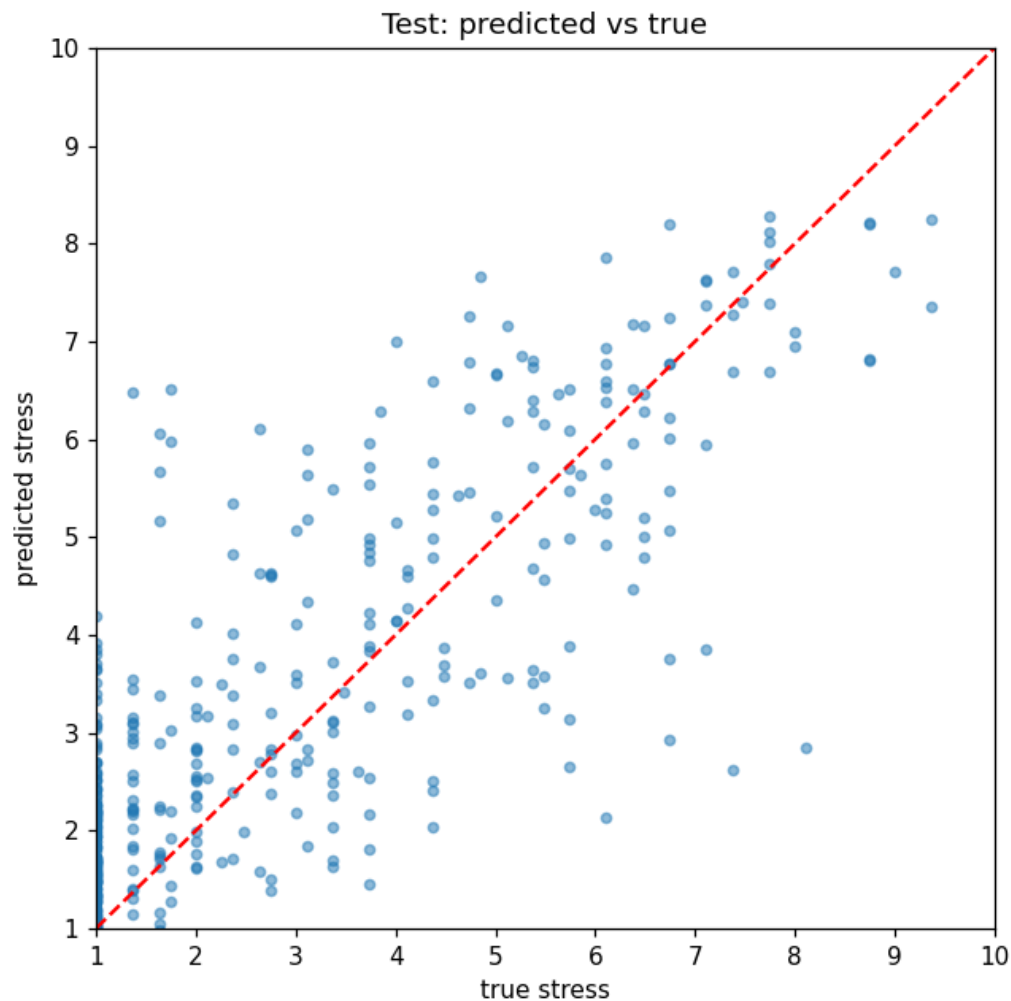


Figure 8.3. Member 2 locked-test continuous predictions versus annotated stress. The red dashed line is the ideal $y=x$ relationship.

Source: exact Member 2 output figure. Predictions preserve overall ordering but compress some extreme values toward the center.

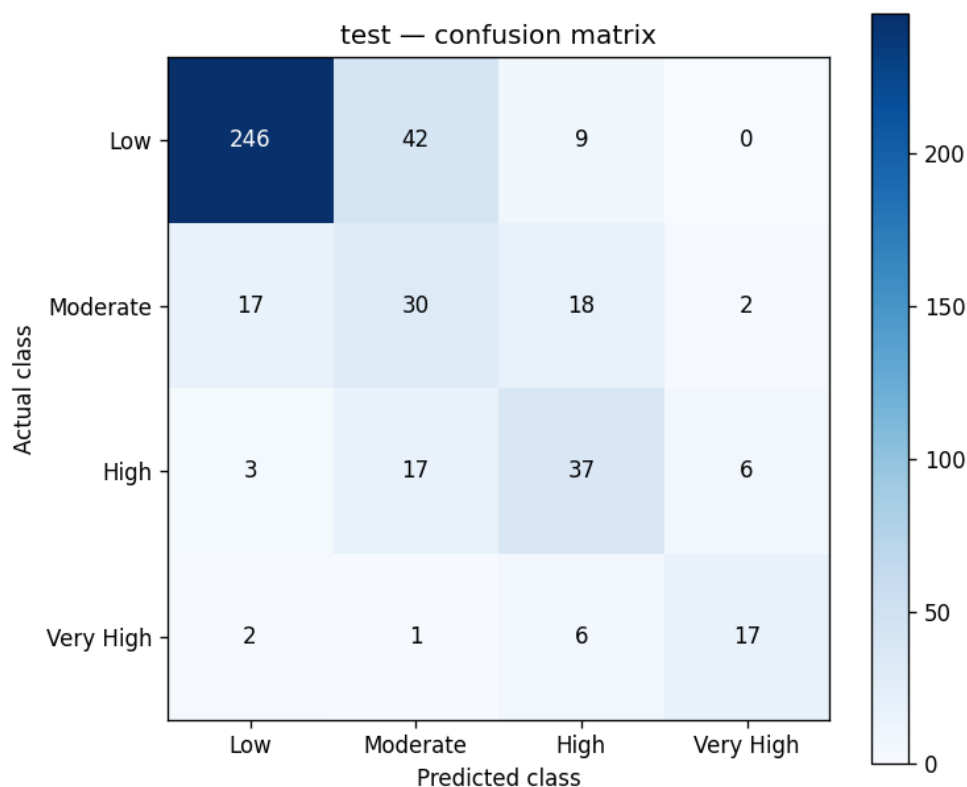


Figure 8.4. Member 2 locked-test confusion matrix using the authoritative Low/Moderate/High/Very High classes.

Source: exact Member 2 output figure. Counts: Low [246,42,9,0], Moderate [17,30,18,2], High [3,17,37,6], Very High [2,1,6,17].

8.6 Explainability, handoff, and limitations

Member 2 explainability is post-hoc and was not used for feature selection. The reproducible package performs residual analysis by stress band and gradient-based token attribution on selected validation examples. Gradient attribution is a local sensitivity diagnostic: it can highlight which token representations most affect one prediction, but it is not causal evidence and can be unstable in contextual Transformer models. The text branch also exhibits regression-to-the-mean at very high annotated scores, visible in the scatter plot and helping explain why raw continuous under-prediction can coexist with a useful ordered ranking.

The primary Member C handoff consists of member2_oof_predictions_deterministic.csv (4,226 rows), member2_validation_predictions_fold_ensemble.csv (452 rows), and member2_test_predictions_fold_ensemble.csv (453 rows), all aligned by unique_post_id. Training rows are genuinely OOF; validation/test prediction is the five-model mean. The fold-model standard deviation is retained as a separate uncertainty diagnostic. CLS embeddings exist only as explicitly labeled exploratory artifacts; the frozen final fusion consumes the scalar Member 2 prediction. Full new-post inference requires the five fine-tuned fold checkpoints; base ParsBERT alone is not a substitute. Because the labeled corpus is AI-assisted and safety-enriched rather than clinically adjudicated or prevalence-sampled, Member 2 outputs are research risk signals, not diagnostic probabilities.

9. Member C: Data Foundation and Learned Fusion

9.1 Duplicate-safe enrichment and connected components

The 5,617 canonical labels were matched back to the cleaned website dataset through duplicate-safe ID/content logic. Two unresolved ambiguous rows were excluded, leaving 5,615 enriched labeled posts. Posts were then connected when they shared an author, a thread, or exact normalized content. The

resulting graph contained 3,622 connected components. Keeping each component inside one role prevents the same author, conversation, or duplicate text from leaking between training and evaluation.

9.2 Final modeling roles and OOF folds

Role	Rows	Purpose
Train	4,226	Fit Member 1, Member 2, and fusion models; divided into five grouped OOF folds
Validation	452	Candidate selection, threshold calibration, and internal-recall constraint evaluation
Locked test	453	One-time final evaluation after model and thresholds were frozen
Embargo	484	Rows withheld because component/source constraints made them unsuitable for clean evaluation

OOF fold	Held-out training rows
0	844
1	844
2	848
3	847
4	843

For each fold, Member 1 and Member 2 trained on the other four folds and predicted the held-out fold. Concatenating the held-out predictions produced one realistic prediction per training row from a model that had not seen that row. These OOF predictions were used as fusion features. Validation and test rows were predicted by averaging the five base-model fold predictions.

9.3 Fusion matrix

The final fusion was learned late fusion, not voting. The fusion table joined records by `unique_post_id` and included Member 1 prediction, Member 2 prediction, their arithmetic mean, signed difference, absolute disagreement, and the engineered tabular features available from Member 1. The final fusion did not ingest the high-dimensional Member 2 text embeddings. This reduced model size, simplified reproduction, and limited overfitting while preserving the strongest text signal through the Transformer score.

Fusion feature sketch

Final stress = CatBoost(M1 prediction, M2 prediction, base mean, M2-M1 difference, |M2-M1|, tabular features). The model can learn conditional nonlinear weighting rather than applying the same average to every post.

9.4 Candidate models and selection policy

Candidate families included each base model alone, fixed weighted blends, ordinary linear stacking, Ridge stacking, scalar CatBoost models using only base scores, and hybrid CatBoost models using scores plus tabular context. Threshold search was performed separately for each candidate. Only candidates with at least one threshold triple satisfying all validation recall objectives were considered feasible.

Among feasible candidates, the selection rule prioritized Very-High precision, then macro F1, then High precision, and finally lower continuous MAE. This explains why the selected model was not simply the candidate with the smallest raw validation MAE. If a non-model blend won, the implementation required fallback to the best feasible learned model under the same ranking.

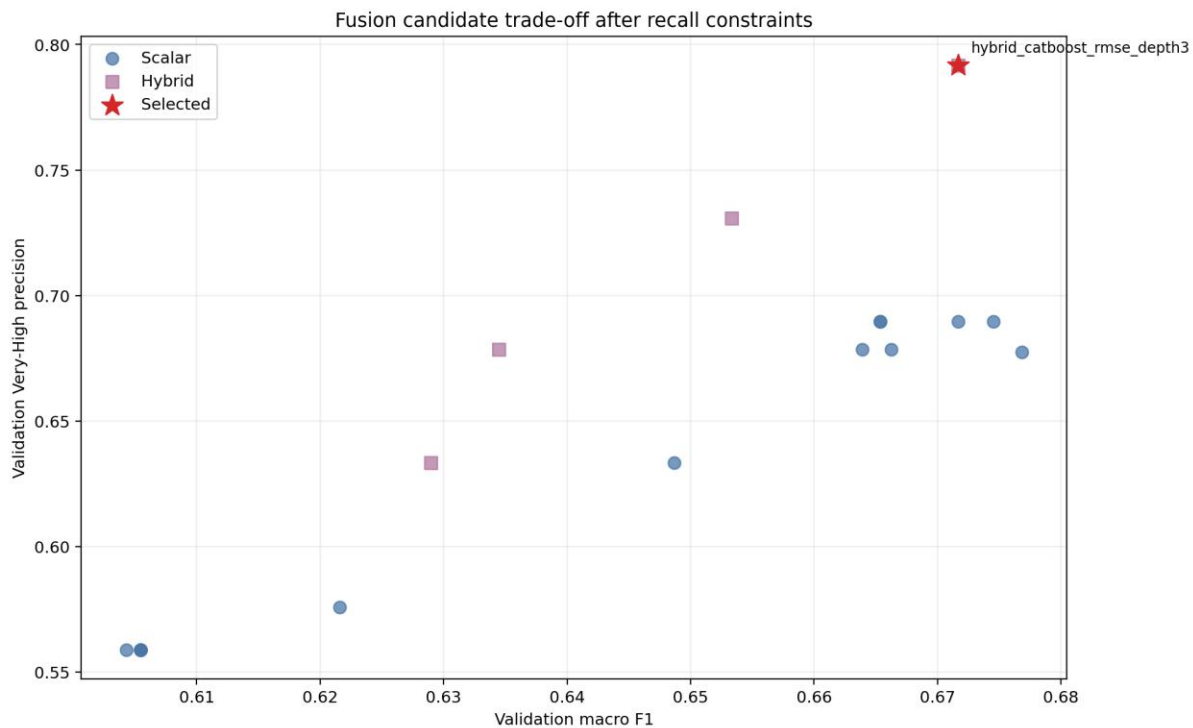


Figure 8. Validation trade-off among fusion candidates. The selected model reflects the constrained ranking rather than a single-metric minimum.

Source: Member C reproducible project, outputs/report_figures/fusion_candidate_tradeoff.png.

9.5 Selected model and thresholds

Setting	Final value
Selected candidate	hybrid_catboost_rmse_depth3
Iterations	450
Tree depth	3
Learning rate	0.035
Loss	RMSE
L2 leaf regularization	10
Random strength	0.5
Bootstrap	Bayesian; bagging temperature 0.5
Score clipping	1.0 to 10.0
Low / Moderate threshold	3.0954385
Moderate / High threshold	4.9051608
High / Very-High threshold	6.4621351

The class thresholds were calibrated only on validation predictions. The true-label definitions remained fixed at 3, 5, and 7; the calibrated prediction thresholds differ because model scores are not perfectly calibrated to the true scale.

10. Experimental Design and Evaluation Protocol

10.1 Why a locked test was used

The validation set was used repeatedly to compare candidate architectures and calibrate thresholds. The test set was therefore locked until the final candidate, feature set, and thresholds were frozen. The evaluation script required explicit acknowledgement, wrote model and prediction hashes, and normally refused a second evaluation unless the user requested a reproducibility-only force run. Rerunning the exact frozen artifacts is acceptable for verification; changing the model or thresholds in response to test results would invalidate the test as untouched evidence.

10.2 Metrics

- Continuous: MAE, RMSE, Pearson correlation, Spearman correlation, and prediction range.
- Operational classes: accuracy, macro F1, per-class precision, recall, F1, support, and confusion matrices.
- Uncertainty: 95% Wilson intervals for per-class recall.
- Safety emphasis: Very-High precision/recall and validation threshold-sensitivity analysis.

Internal recall objectives were Low $\geq 75\%$, Moderate $\geq 50\%$, High $\geq 50\%$, and Very High $\geq 75\%$. These were self-defined project objectives, not clinical, regulatory, or universal deployment requirements. Passing or failing them is reported transparently but is not treated as an automatic deployment decision.

10.3 Sample weighting

Training weights combined provisional annotation-source reliability and square-root inverse-frequency class balancing. The final class-weight multipliers were approximately 0.684 for Low, 1.056 for Moderate, 1.198 for High, and 1.925 for Very High, normalized to a mean sample weight of 1.0. Validation and test metrics were not reweighted.

11. Results

11.1 Continuous prediction performance

Dataset	MAE	RMSE	Pearson r	Spearman rho
Fusion validation	0.8348	1.1336	0.8577	0.8185
Fusion locked test	0.9174	1.2459	0.8333	0.7901
Member 1 validation	1.2657	1.5924	0.6886	0.6577
Member 1 locked test	1.2889	1.6556	0.6727	0.6261
Member 2 validation	0.8246	1.1462	0.8596	0.8061
Member 2 locked test	0.9689	1.3182	0.8251	0.7891

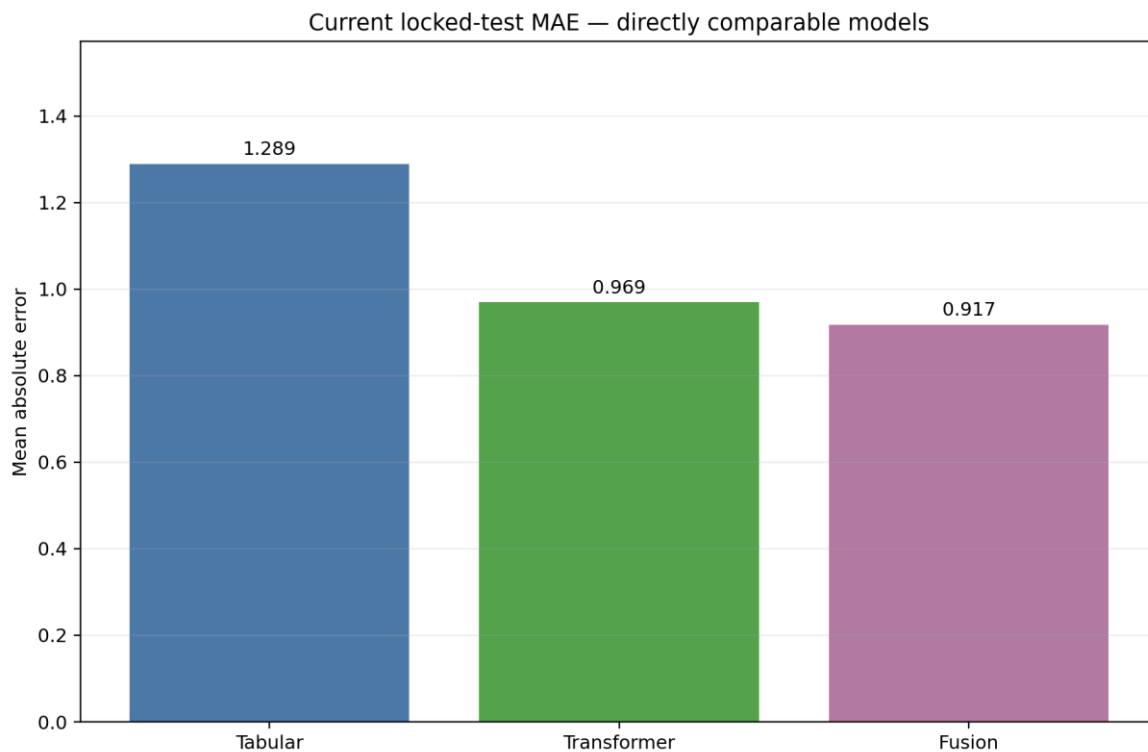


Figure 9. Same locked-test MAE for Member 1, Member 2, and the final fusion. The fusion improves over both individual branches.

Source: Member C reproducible project. Values: tabular 1.2889, Transformer 0.9689, fusion 0.9174.

On the same 453-post locked test, Member 1 obtained MAE 1.2889, Member 2 obtained MAE 0.9689, and the final fusion obtained MAE 0.9174. The fusion therefore reduced locked-test MAE by approximately 5.3% relative to the Transformer and 28.8% relative to the tabular branch. Because all three values use the same frozen test rows, this is the cleanest internal model comparison in the project. The result supports the late-fusion design: the Transformer supplies the strongest single branch, while structured signals still improve the final out-of-sample prediction.

11.2 Validation and locked-test classification results

Class	Validation precision	Validation recall	Test precision	Test recall	Internal recall target
Low	0.9194	0.8451	0.9155	0.8754	0.75
Moderate	0.3824	0.5821	0.3544	0.4179	0.50
High	0.6226	0.5238	0.5690	0.5238	0.50
Very High	0.7917	0.7600	0.6250	0.7692	0.75

Metric	Validation	Locked test
Accuracy	0.7566	0.7528
Macro F1	0.6717	0.6284
All four internal recall objectives passed?	Yes	No - Moderate missed

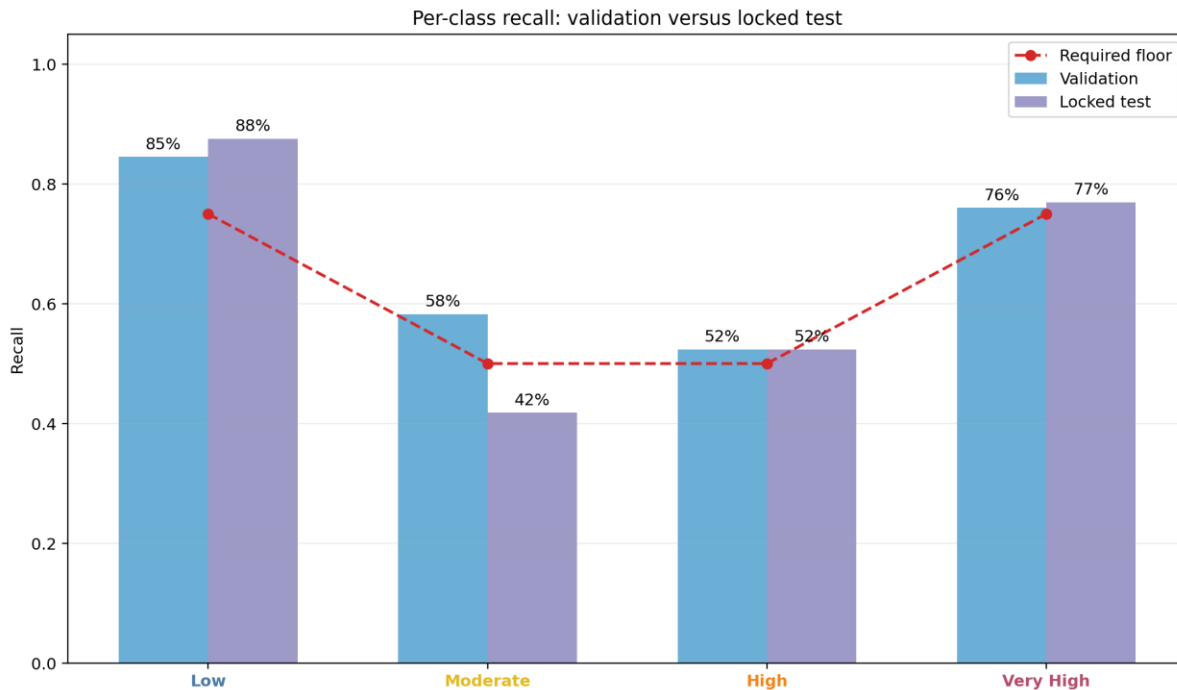


Figure 10. Per-class recall on validation and the locked test against the project-defined recall floors.

Source: Member C reproducible project. Validation passed all four internal objectives; the locked test missed only Moderate.

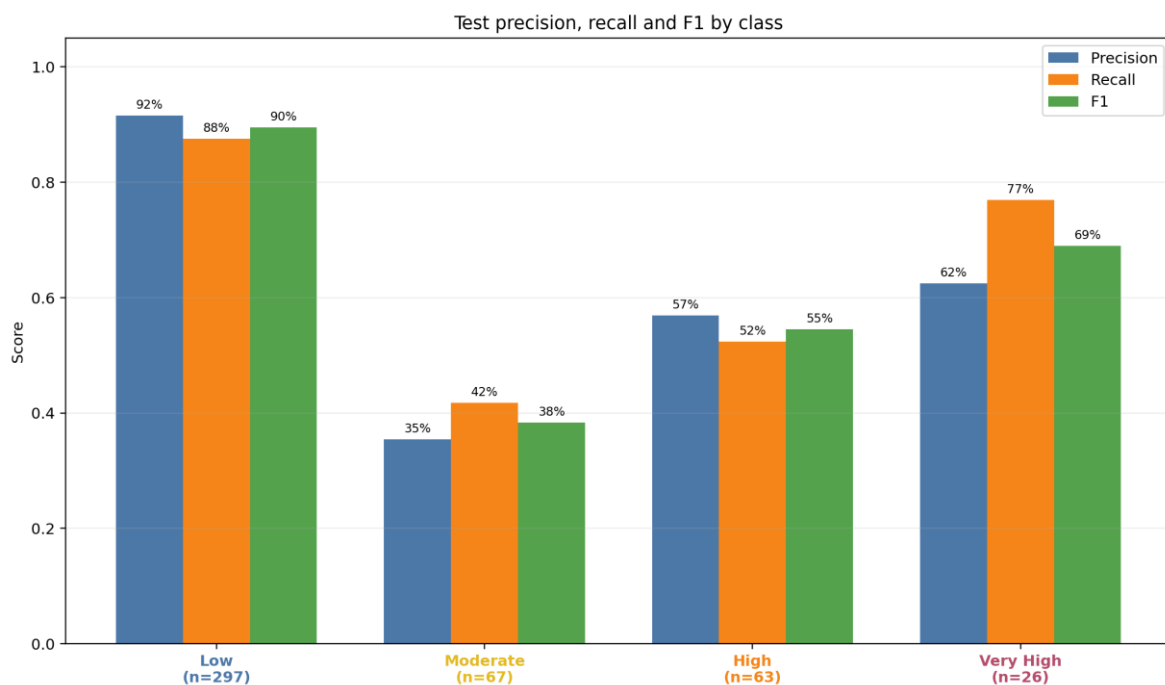


Figure 11. Locked-test precision, recall, and F1 by operational class.

Source: Member C reproducible project, outputs/report_figures/test_precision_recall_f1.png.

11.3 Score distributions and confusion structure

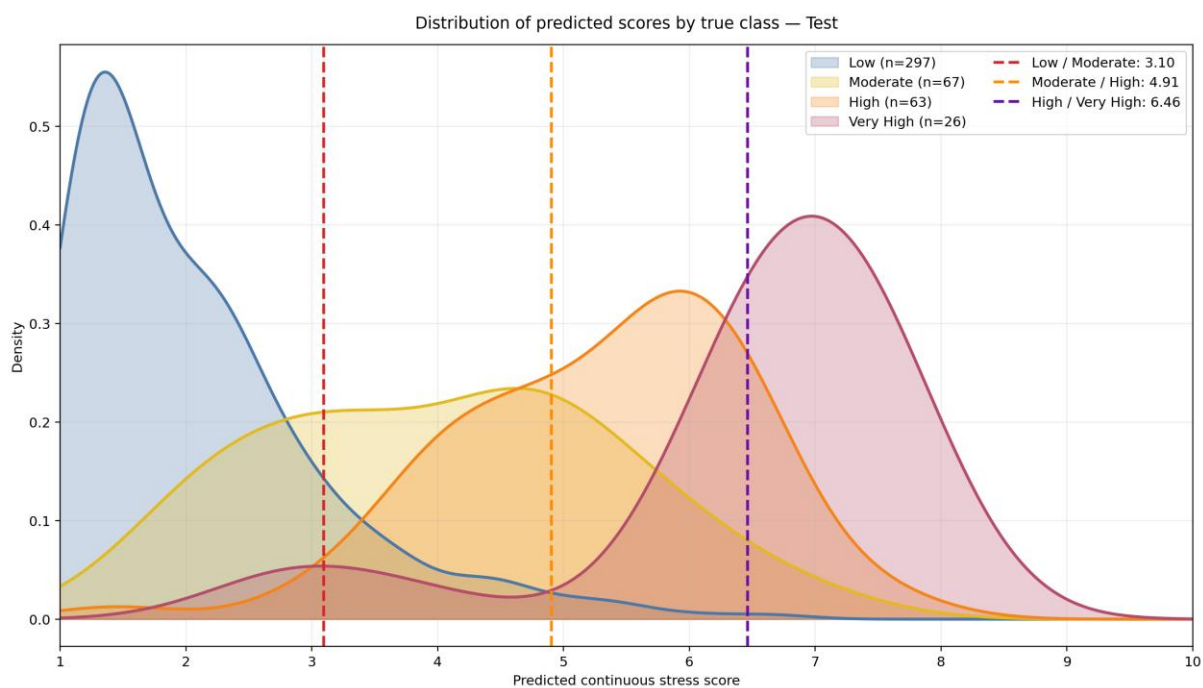


Figure 12. Locked-test fusion-score distributions grouped by authoritative true class. The vertical lines are validation-calibrated prediction thresholds.

Source: Member C reproducible project. Overlap around the first two thresholds explains much of the Moderate error.

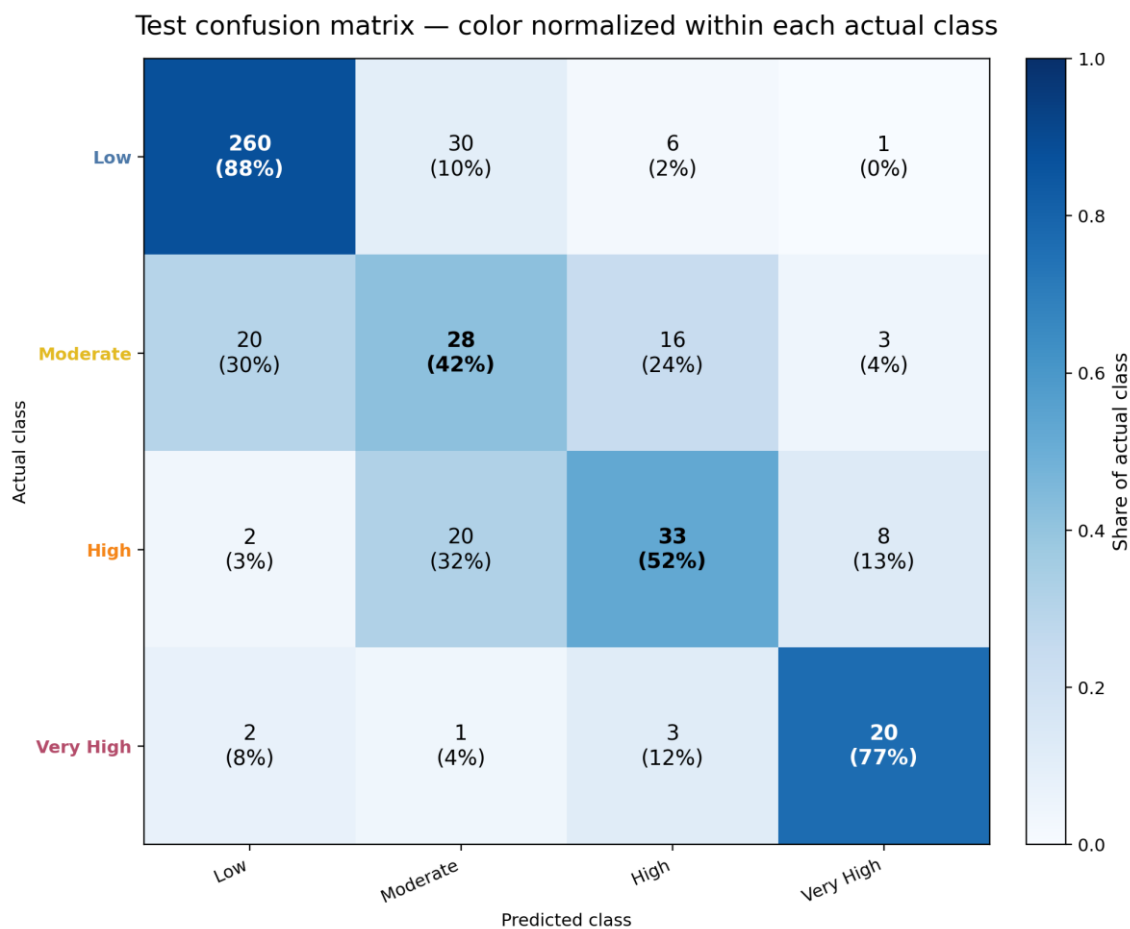


Figure 13. Row-normalized locked-test confusion matrix. Each row shows where posts from one true class were predicted.

Source: Member C reproducible project. Counts: Low [260,30,6,1], Moderate [20,28,16,3], High [2,20,33,8], Very High [2,1,3,20].

Moderate was not ignored completely; its errors were split in both directions. Of 67 true Moderate posts, 20 were predicted Low, 28 Moderate, 16 High, and 3 Very High. This indicates overlap around both Moderate boundaries rather than a simple one-sided threshold error. In contrast, 20 of 26 Very-High posts were detected, meeting the primary internal safety objective.

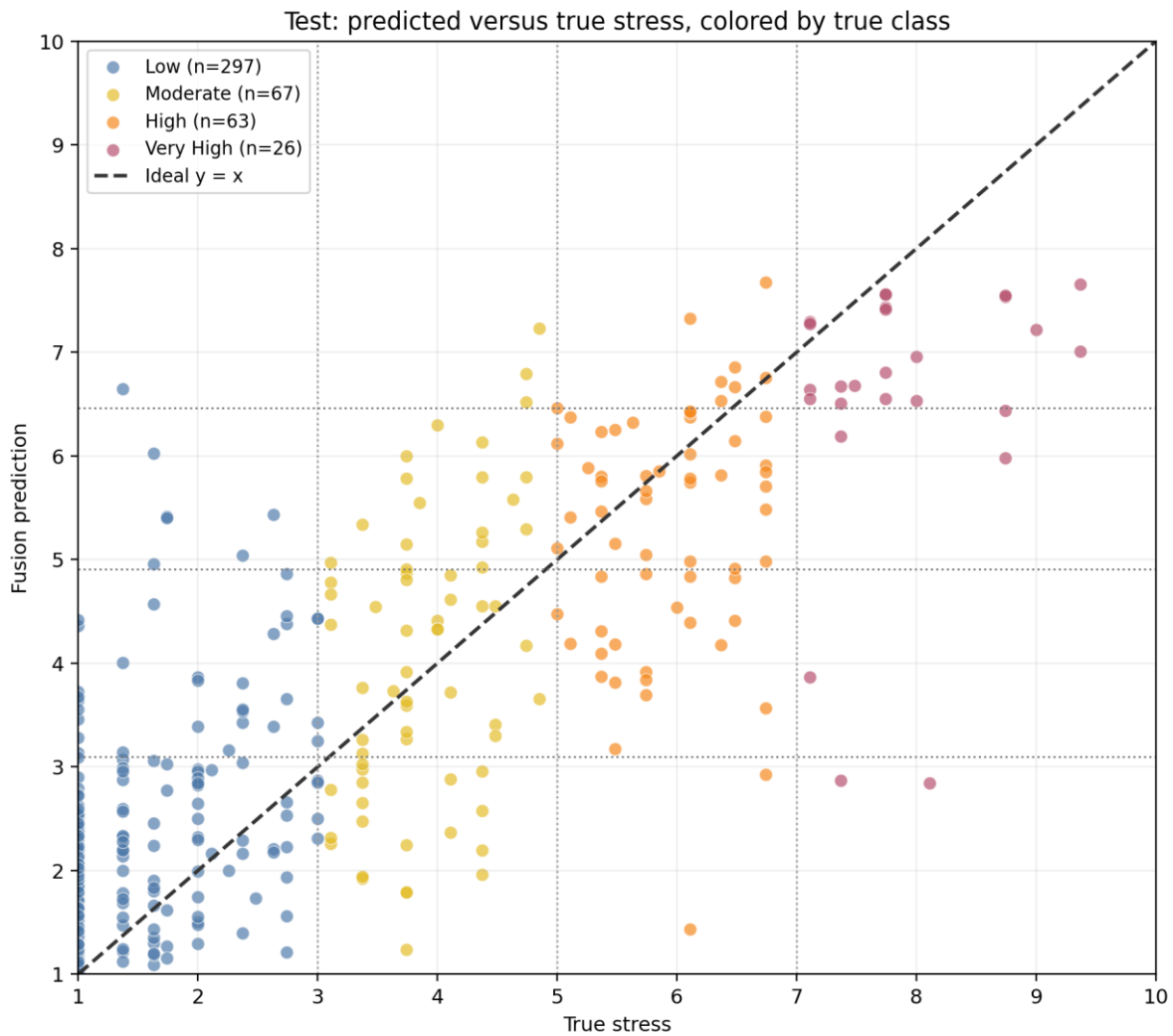


Figure 14. Locked-test continuous predictions against true stress, colored by true class. Vertical lines show true-score boundaries and horizontal lines show calibrated prediction thresholds.

Source: Member C reproducible project, outputs/report_figures/test_true_vs_predicted_by_class.png.

11.4 Interpretation of project acceptance

Final project interpretation

The model met three of four internal recall objectives on the locked test and met the principal Very-High objective. Moderate detection remains an improvement area. The model is therefore best described as a candidate for a human-in-the-loop risk-monitoring overlay, not as an autonomous diagnostic system and not as “unusable” solely because one self-defined target was missed.

12. Explainability and Error Analysis

12.1 Global feature importance

CatBoost feature importance and validation mean absolute SHAP were calculated after training. They were used to explain and audit the selected model, not as the formal feature-selection procedure. SHAP assigns each feature a local contribution relative to the model baseline; mean absolute SHAP summarizes contribution magnitude across many rows [13]. Correlated features can share or redistribute importance, so SHAP should not be interpreted as causality.

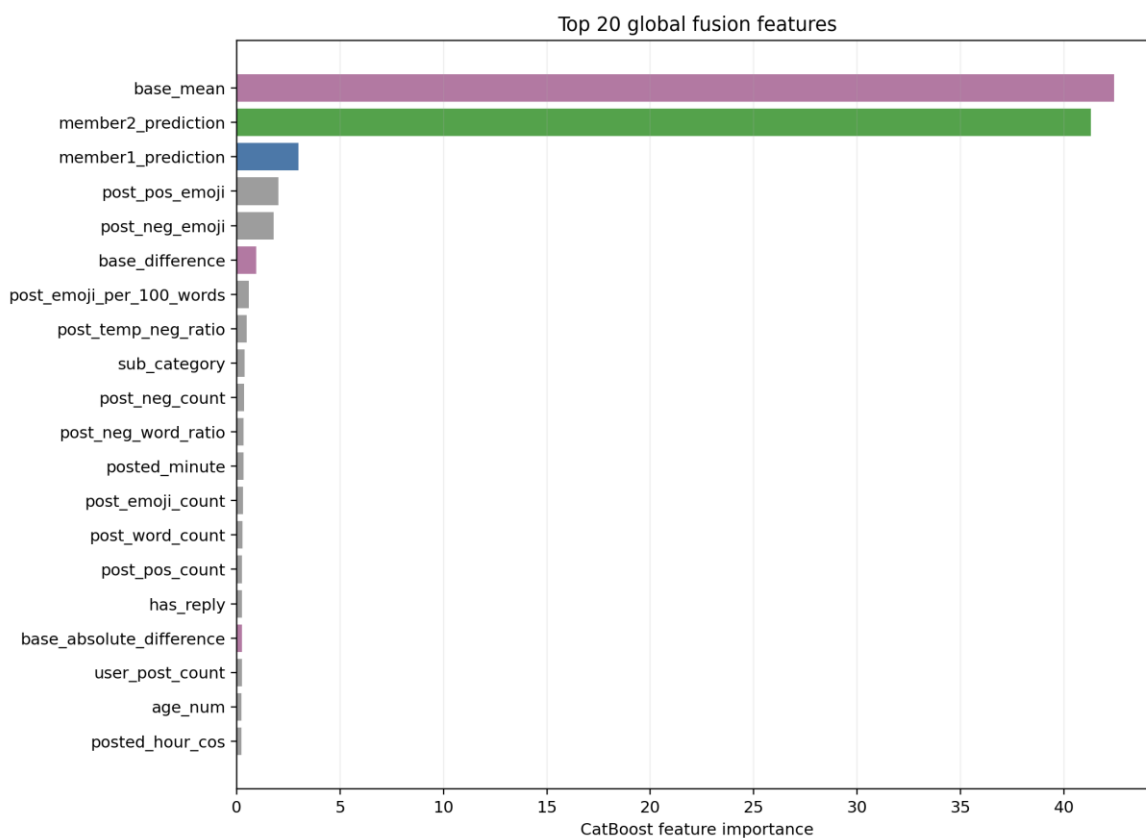


Figure 15. Top global CatBoost feature importances for the selected hybrid fusion.

Source: Member C reproducible project. The base-model mean and Member 2 prediction dominate the split-based importance.

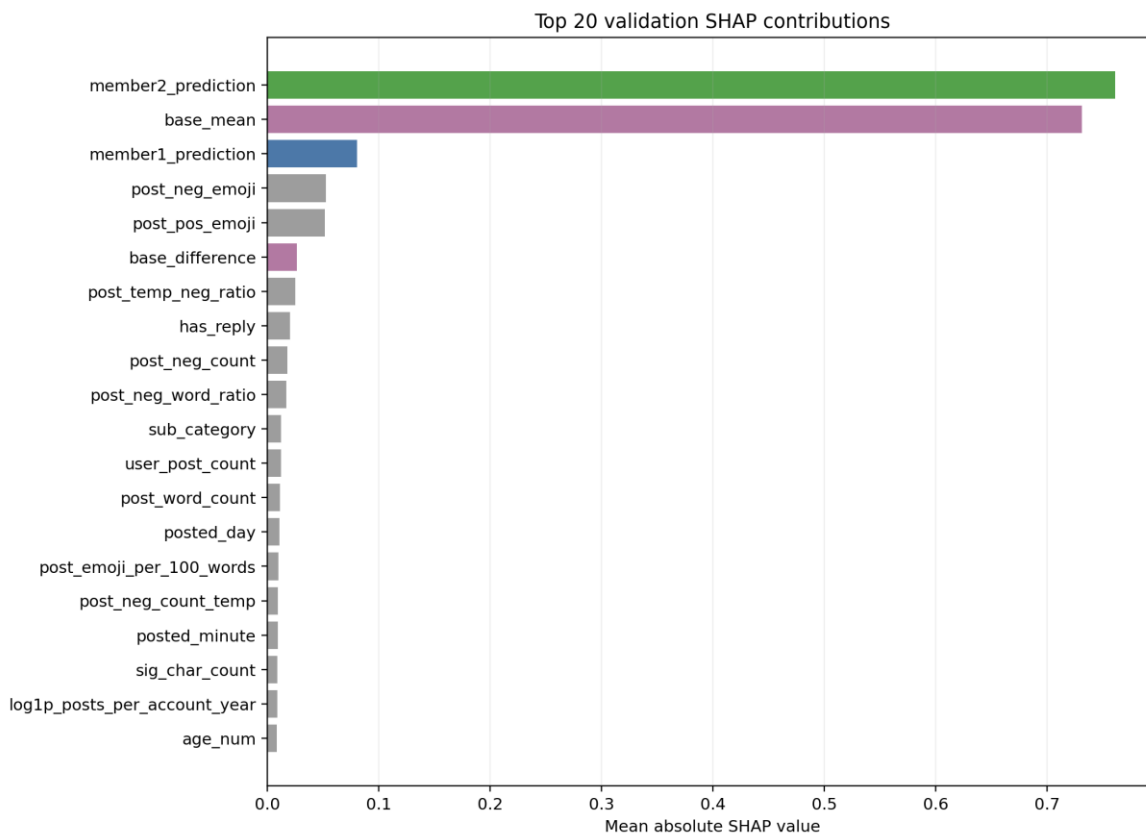


Figure 16. Top validation mean absolute SHAP contributions for the selected hybrid fusion.

Source: Member C reproducible project. Member 2 prediction and base mean dominate, followed by the Member 1 score and selected emoji/lexical/context features.

The strongest signals were Member 2 prediction and base_mean. Member 1 prediction remained useful but much smaller globally, while negative/positive emoji counts, model disagreement, negative-word ratios, sub-category, reply status, and posting features supplied complementary context. This supports the chosen late-fusion interpretation: semantic text prediction carries most of the signal, but structured features improve calibration and difficult cases.

For Member 1 specifically, the reproducible tabular package generated global CatBoost feature importance averaged across the five fold models. The leading features were is_starter, post_neg_emoji, post_char_count, post_neg_count, post_pos_word_ratio, post_temp_neg_ratio, post_word_count, and sub_category. This analysis was performed after model fitting for interpretation and audit only; it did not drive formal feature removal or model selection. Native CatBoost SHAP support exists in the source code, but the final executed Member 1 artifact recorded shap_run=false, so no Member 1 SHAP result is claimed.

12.2 Threshold and error trade-offs

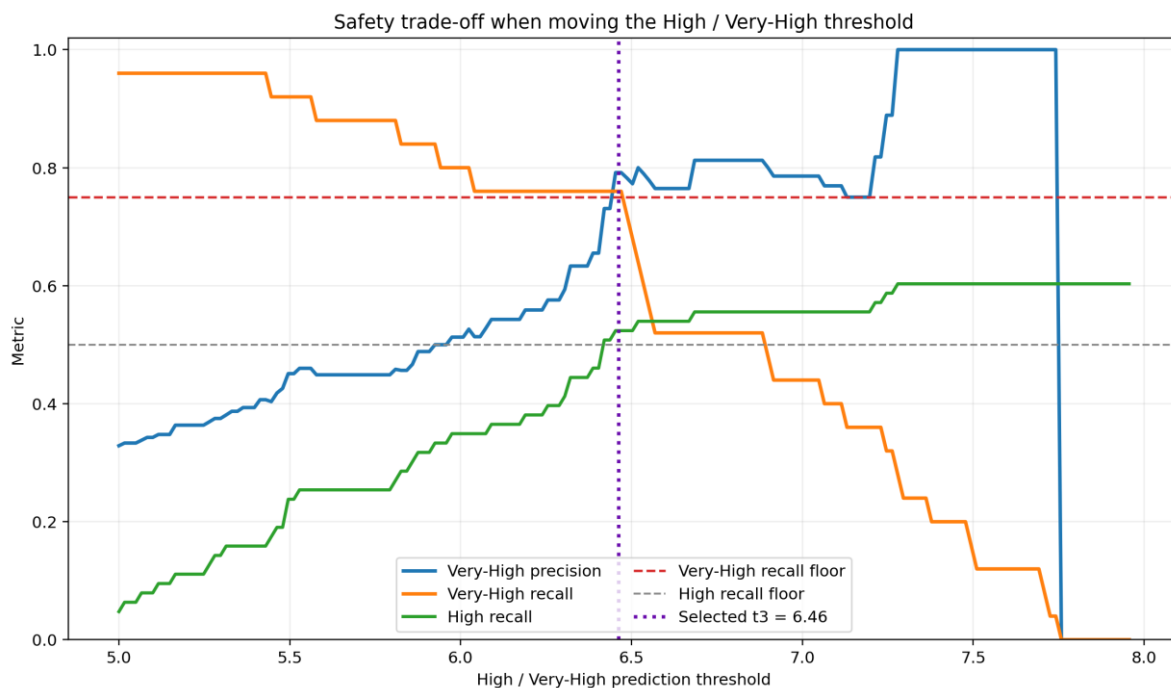


Figure 17. Validation sensitivity of Very-High precision, Very-High recall, and High recall to the upper threshold while holding the first two thresholds fixed.

Source: Member C reproducible project. The selected threshold balances the project objectives rather than maximizing a single metric.

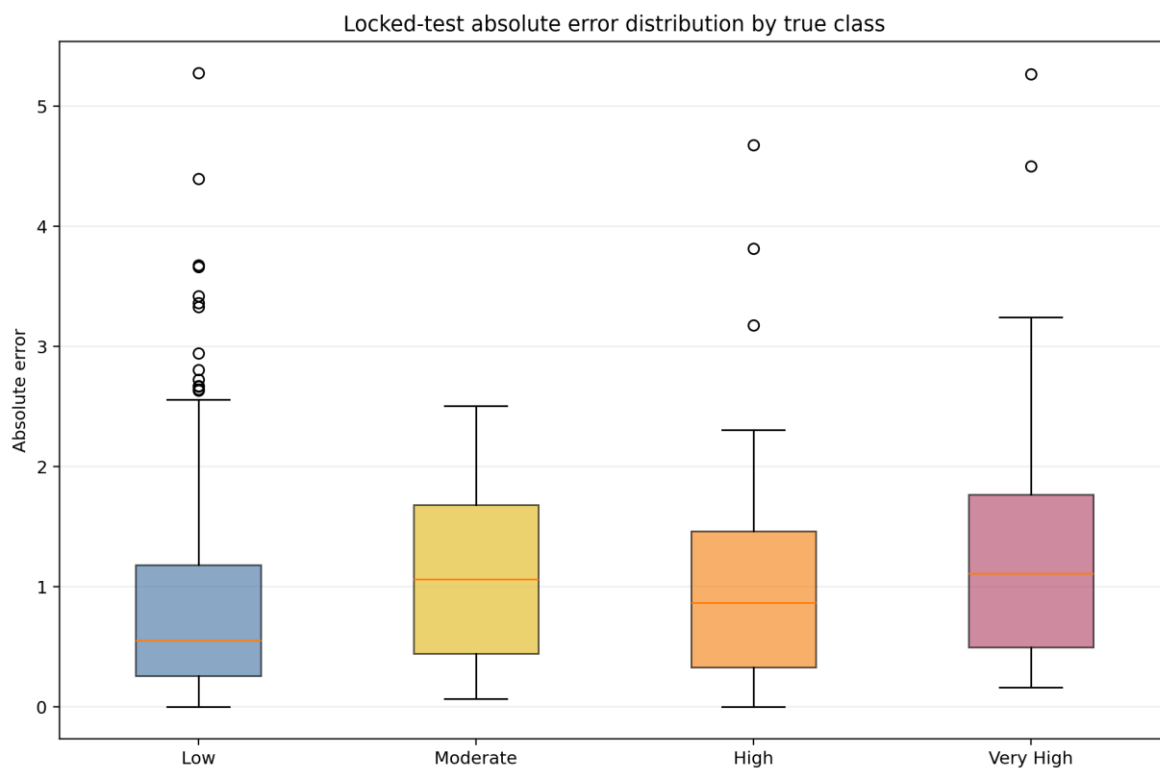


Figure 18. Distribution of absolute locked-test regression error by true class.

Source: Member C reproducible project, outputs/report_figures/test_absolute_error_by_true_class.png.



Figure 19. Locked-test underprediction and overprediction patterns by true class.

Source: Member C reproducible project, outputs/report_figures/test_error_direction_by_true_class.png.

13. Deployment Interpretation, Ethics, and Privacy

13.1 Proposed human-in-the-loop overlay

A realistic deployment would run beside the website rather than replace moderators or clinicians. For each new post, the tabular branch and Persian text branch produce continuous scores, the fusion produces a final score and operational class, and the result updates a private per-user temporal state. A trained reviewer receives prioritized cases rather than public labels.

18. Generate Member 1 and Member 2 predictions for the new post.
19. Generate the fused continuous stress score and operational class.
20. Update time-decayed mean stress, recent maximum, posting frequency, repeated High/Very-High count, disagreement, and change from the user's baseline.
21. Apply approved immediate, sustained, repeated, and escalating-risk rules.
22. Place qualifying users into a private moderator or qualified-support review queue.
23. Allow the human reviewer to inspect recent context and decide whether supportive resources, professional review, or another proportionate response is appropriate.

Temporal aggregation can partly mitigate weak Moderate classification. A user whose individual posts move between Low, Moderate, and High can still be surfaced when the trajectory is sustained or rising. Conversely, isolated uncertain posts need not automatically trigger high-intensity action.

13.2 Ethical and governance requirements

Risk	Required control
Privacy and re-identification	Minimize stored identifiers, restrict access, encrypt data, separate research IDs from platform identities, and avoid publishing raw text.
Consent and lawful basis	Obtain institutional/legal review for any operational monitoring; clearly define whether and how users are informed.
False positives	Use private human review and proportional responses; never publicly label users.
False negatives	Do not present the system as comprehensive screening; preserve ordinary moderation and access to support.
Bias and representativeness	Audit performance across available subgroups and categories; do not infer unobserved demographics.
Automation overreach	No autonomous diagnosis, emergency referral, or medical action.
LLM-generated labels	Seek expert validation and compare with validated instruments before clinical claims.
Auditability	Log model version, threshold version, alert reason, reviewer action, and outcome where legally and ethically permitted.

13.3 Recommended rollout stages

24. Historical replay on de-identified data to measure alert volume and identify failure modes.
25. Silent monitoring with no user-facing action, reviewed only by the research/moderation team.
26. Limited moderator pilot with defined escalation criteria and outcome logging.
27. Evaluation of false-alert burden, missed severe cases, reviewer agreement, and subgroup behavior.
28. Controlled supportive user-facing workflow only after privacy, consent, training, and governance approval.
29. Formal clinical or crisis-service partnership before any medical/emergency integration.

14. Limitations

- Ground truth is AI-assisted annotation, not diagnosis or validated clinical screening. The weighting scheme reflects project trust decisions and requires external validation.
- The labeled set is safety-enriched and is not representative of natural website prevalence.
- The targeted literature review did not establish an absolute “first” claim.
- Forum users are self-selected and the project lacks reliable demographic representativeness, pregnancy-stage confirmation, and clinical outcomes.

- Historical missing-gender imputation to female can introduce bias; future pipelines should preserve unknown values.
- Some labels are calibrated single-rater estimates with lower provenance weight.
- The final test is relatively small in Very High (26) and other minority classes, leading to wide recall intervals.
- Moderate remains difficult because it overlaps both adjacent score regions.
- The pre/post active-learning comparison is not causal because dataset composition, split sizes, and leakage controls changed.
- The final fusion depends on the availability and versioning of both upstream branch models for live inference.
- SHAP explains model behavior rather than causality or clinical correctness.
- Forum language and behavior can change over time, requiring drift monitoring and new untouched evaluation data.

15. Conclusions and Future Work

This project demonstrates that it is possible to construct a complete Persian pregnancy-forum stress-risk pipeline from an initially unstructured and unlabeled source. The main contribution is broader than a single model: robust collection, Persian cleaning, proxy-guided sampling, multi-round annotation, active learning, provenance-aware label reconstruction, connected-component leakage prevention, OOF base predictions, constrained learned fusion, locked evaluation, explainability, and reproducible notebook delivery.

The final hybrid fusion produced the best same-test continuous accuracy and met the primary internal Very-High recall objective. It did not meet the aspirational Moderate recall target, and this limitation must remain visible. The correct next step is not autonomous deployment and not further tuning on the opened test. It is a controlled human-in-the-loop evaluation with a new annotation and confirmation cycle.

15.1 Priority future work

30. Conduct expert review of a stratified label subset and compare stress scores with validated perinatal stress/anxiety/depression instruments.
31. Perform qualitative analysis of Moderate-to-Low and Moderate-to-High errors and annotate new boundary cases.
32. Reserve a new untouched dual-annotated confirmation set before claiming improved generalization.
33. Evaluate ordinal objectives such as stress > 3 , > 5 , and > 7 in addition to four-bin classification.
34. Test user-level temporal models with time-decayed averages, sequence models, and calibrated alert rules.
35. Audit performance by category, post type, time, text length, and available non-sensitive subgroup variables.
36. Replace historical gender imputation with an explicit unknown category and perform fairness-oriented sensitivity analysis.
37. Measure moderator workload, alert precision, missed severe cases, and reviewer agreement in offline replay and a limited pilot.
38. Create governance documentation covering consent, privacy, retention, access control, model updates, and escalation.
39. Prepare the dataset and manuscript for expert and ethics review before any public release or peer-reviewed submission.

16. Generative AI Assistance and Author Responsibility

OpenAI ChatGPT and Anthropic Claude materially assisted parts of this project and are therefore disclosed and cited as software-based research assistants [20,21].

Claude and ChatGPT were used as independent AI-assisted annotators during dataset construction. ChatGPT additionally assisted with implementation review, debugging, reproducibility design, documentation, report drafting and editing, and artifact preparation. These tools did not replace the team's design decisions, data engineering, model training, validation, interpretation, or final review. All AI-generated suggestions and labels were treated as fallible project inputs rather than clinical judgments, and the authors retain full responsibility for the submitted work.

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Software and artifact attribution note

All figures labeled as project-generated were produced from the project's scraped/processed data or saved model outputs. Open-source methods and libraries are cited above. Generative-AI assistance is disclosed in Section 16 and References [20,21].
Project repository: <https://github.com/MohamadRezaDarvish/stress-detection-persian-forum>

Appendix A. Additional Validation and Diagnostic Figures

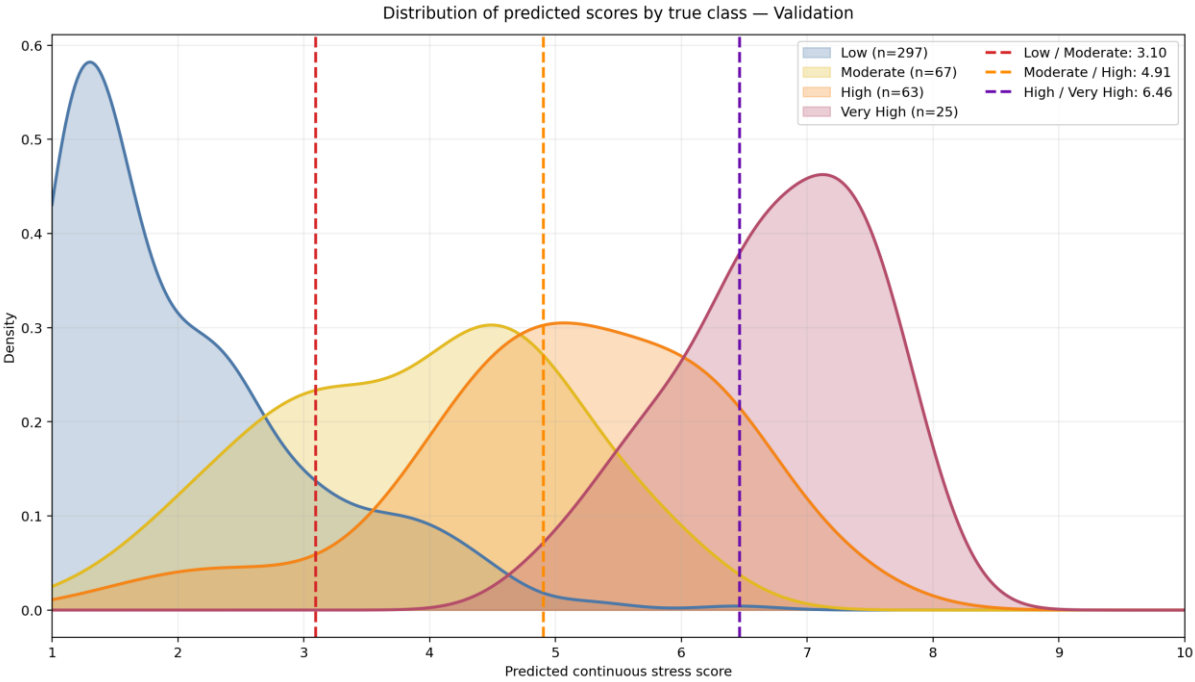


Figure 20. Validation fusion-score distributions with calibrated thresholds.

Source: Member C reproducible project.

Validation confusion matrix — color normalized within each actual class

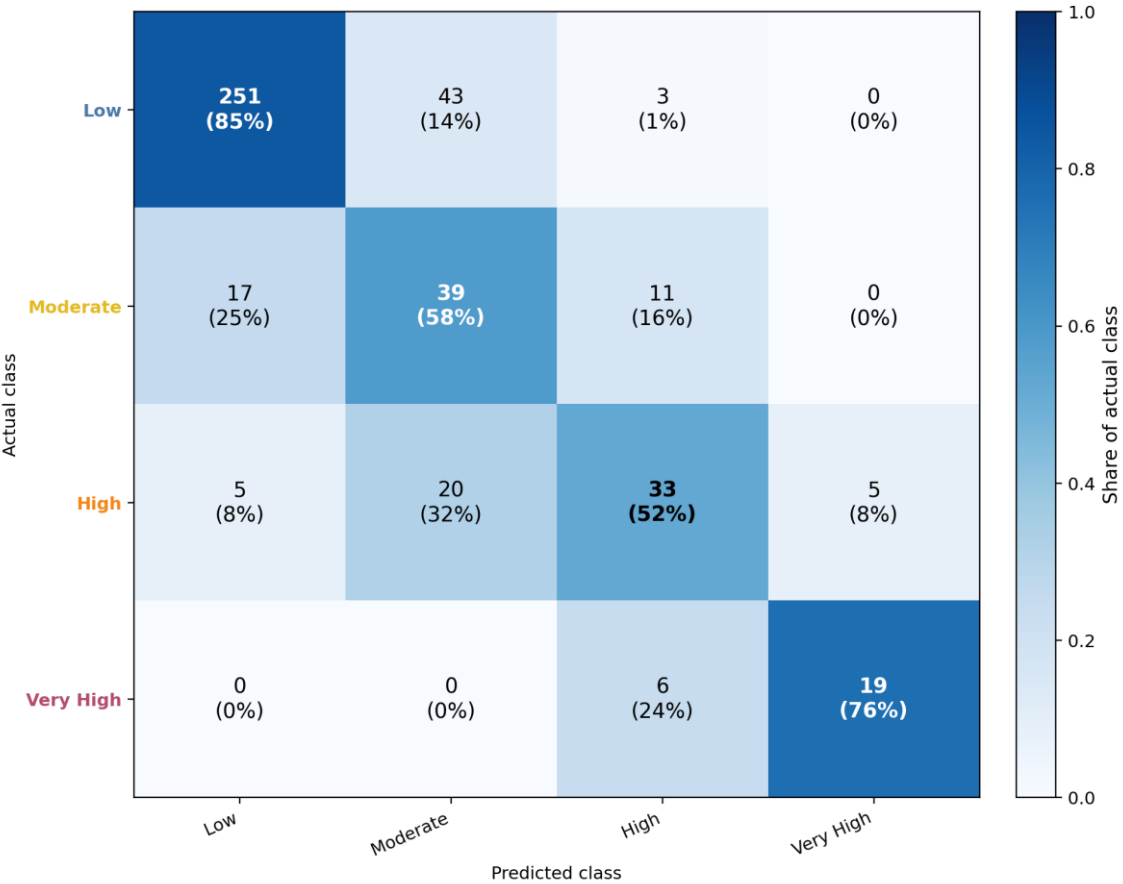


Figure 21. Row-normalized validation confusion matrix.

Source: Member C reproducible project.

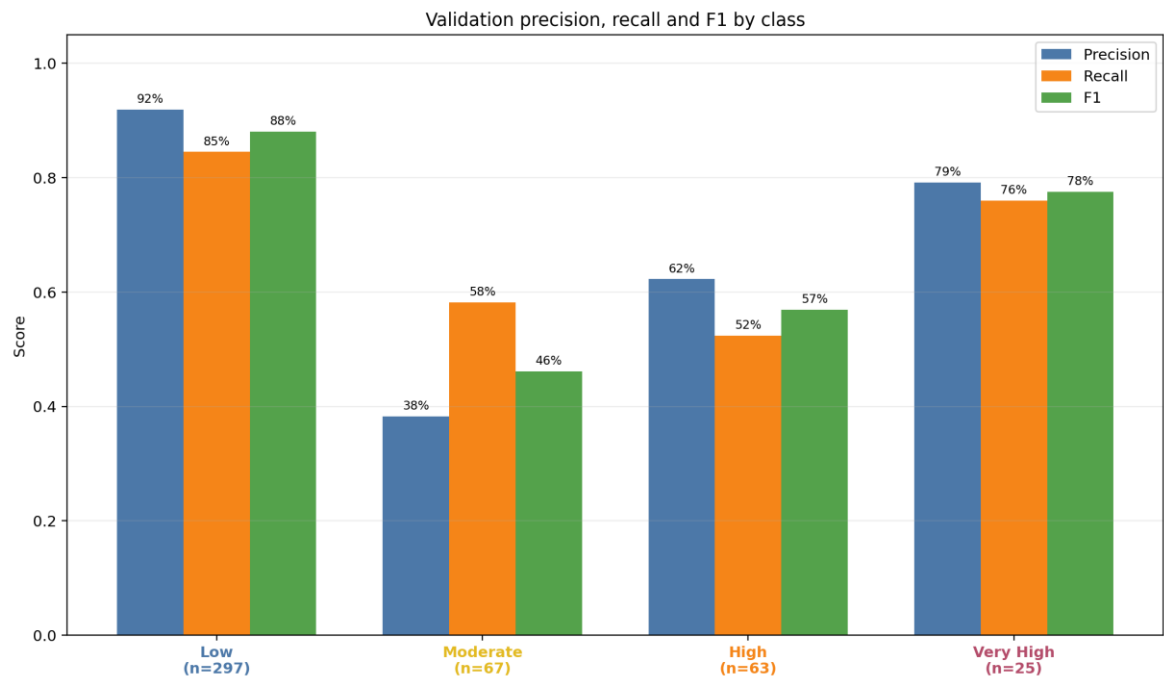


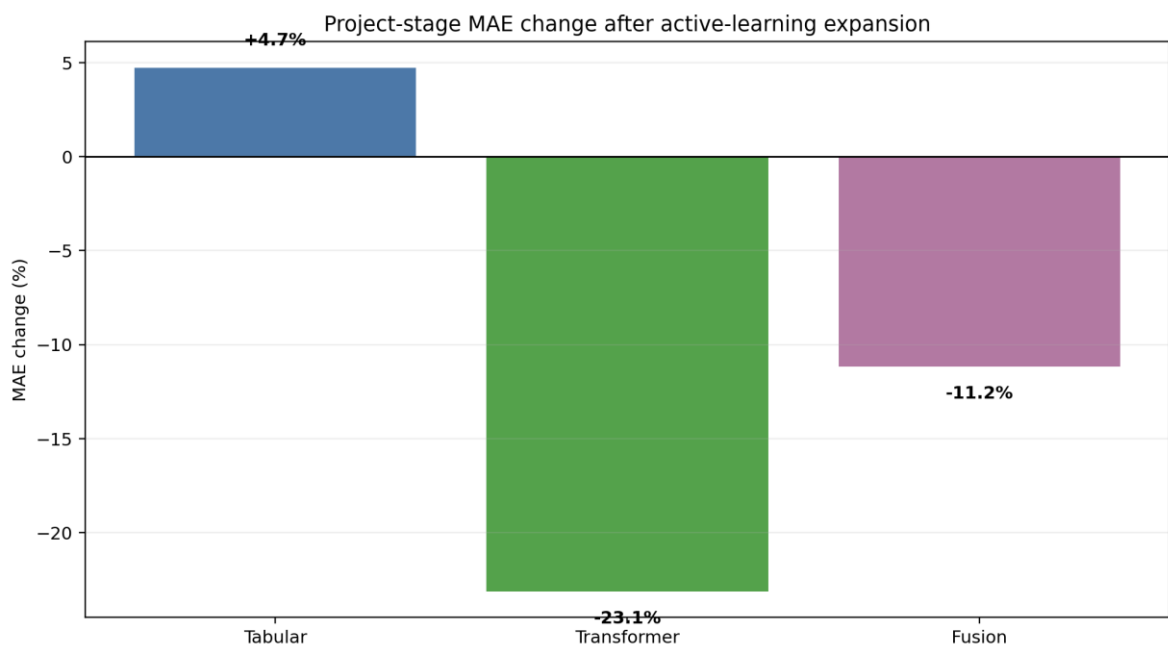
Figure 22. Validation precision, recall, and F1 by class.

Source: Member C reproducible project.



Figure 23. Validation true-versus-predicted continuous stress, colored by true class.

Source: Member C reproducible project.



Negative change means lower MAE. Historical pre-active-learning and current post-active-learning values use different split sizes, label pools and leakage controls. The comparison shows project-stage change, not a controlled causal estimate of active learning.

Figure 24. Percentage change in MAE from the historical pre-active stage to the current stage. Negative values indicate lower error.

Source: Member C reproducible project; descriptive comparison only.

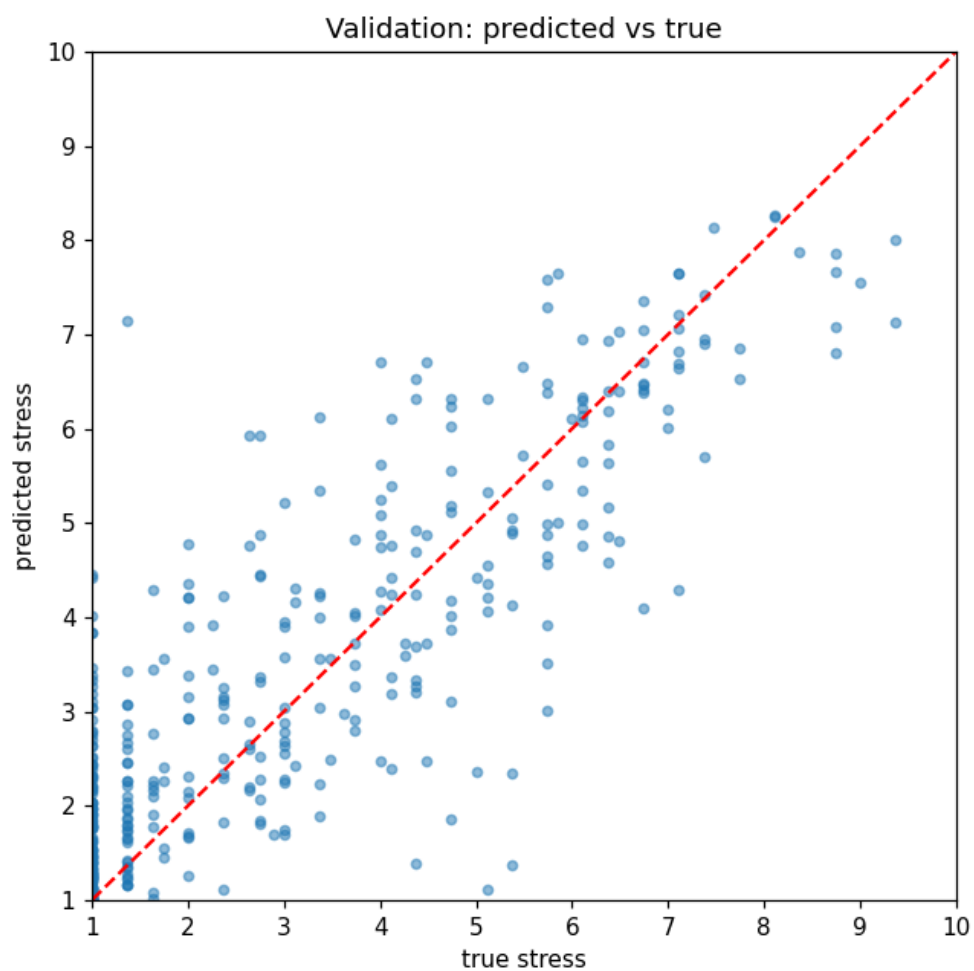


Figure 25. Member 2 validation continuous predictions versus annotated stress.

Source: exact Member 2 validation output figure.

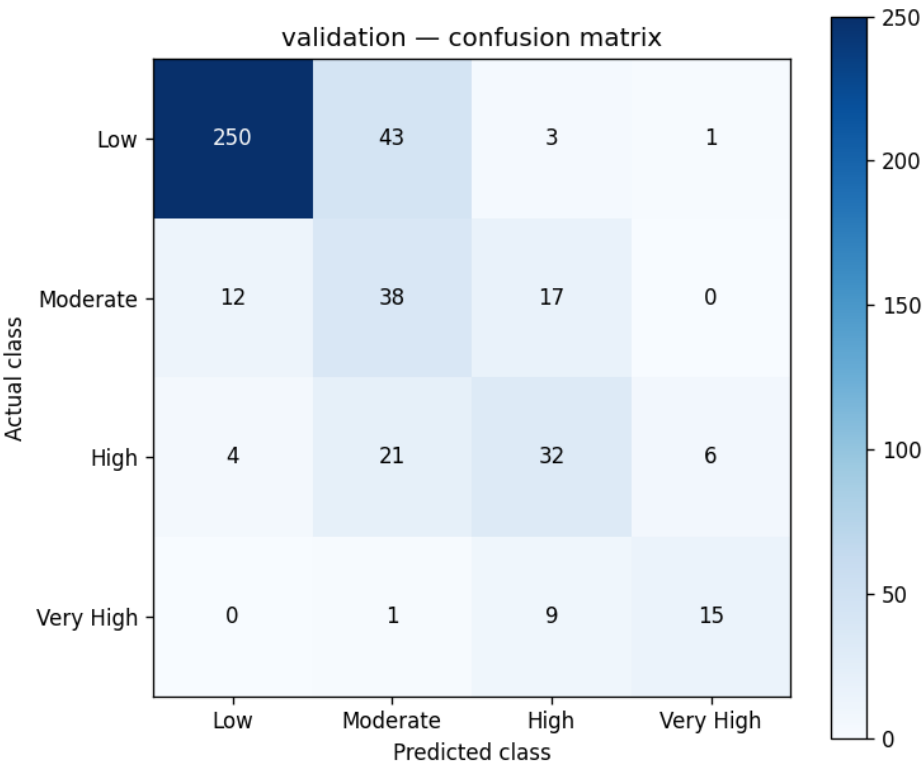


Figure 26. Member 2 validation confusion matrix.

Source: exact Member 2 validation output figure. Counts: Low [250,43,3,1], Moderate [12,38,17,0], High [4,21,32,6], Very High [0,1,9,15].

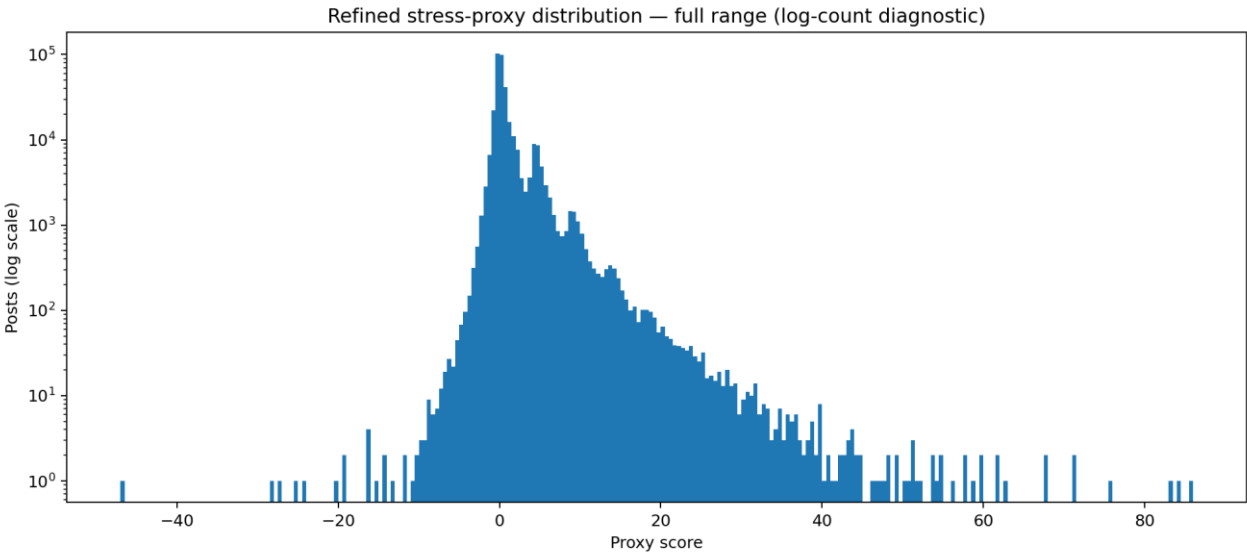


Figure 27. Full-range refined stress-proxy distribution on a log-count axis. The log scale reveals the rare negative and high-positive tails that are visually compressed in a conventional count histogram.

Source: Project-generated from the finalized historical-faithful stress-proxy reconstruction; the figure is diagnostic and the proxy remains a sampling heuristic, not the supervised stress target.

Appendix B. Reproducibility and Artifact Map

Artifact group	
Member C Phases 1-3 notebook	Canonical label reconstruction, calibration, duplicate audits, raw enrichment/cache, connected components, final roles, OOF folds, weights, handoffs, artifact validation and invariants.

Artifact group	Primary content
Member C Phases 4-8 notebook	Handoff validation, candidate training, constrained threshold calibration, fusion OOF, locked-test reproduction, SHAP, prediction smoke test, temporal alerts, active-learning demo, figures, artifact verification.
Final model	models/fusion_selected.cbm and models/fusion_bundle.json
Final metrics	outputs/final_metrics.json, validation_metrics.json, test_metrics.json, confusion matrices, per-class tables
Prediction evidence	fusion_oof_predictions.csv, validation_predictions.csv, test_predictions.csv
Explainability	global_feature_importance.csv, validation_shap_summary.csv, example_fusion_shap.csv
Operational prototypes	temporal_alert_demo.csv and active_learning_demo_top50.csv
Figures	outputs/report_figures/*.png and *.svg
Member 1 main notebook	notebooks/Member1_Full_Reproduction.ipynb and Member1_Full_Reproduction_executed.ipynb; single entry point for validation, feature engineering, five-fold training/loading, OOF/ensemble generation, evaluation, explainability, handoff export and tests.
Member 1 source and configuration	src/ feature engineering, validation, fold, training, prediction, evaluation, explainability, export and verification modules; configs/project_config.json and configs/model_config.json; data/processed/member1_exact_feature_order.csv and member1_preprocessor.joblib.
Member 1 fold models	models/fold_0/model.cbm through models/fold_4/model.cbm; models/final_full_train/model.cbm archived separately.
Member 1 prediction handoff	data/handoff/member1_oof_predictions.csv; member1_validation_predictions.csv; member1_test_predictions.csv; exact alignment key unique_post_id and prediction column prediction.
Member 1 engineered-feature handoff	data/handoff/member1_train_engineered_features.csv.gz; member1_validation_engineered_features.csv.gz; member1_test_engineered_features.csv.gz; data/handoff/member1_handoff_manifest.json.
Member 1 metrics and explainability	outputs/metrics/member1_metrics.json and per-class/confusion-matrix files; outputs/explainability/catboost_global_feature_importance.csv; outputs/figures/global_feature_importance.png and true-vs-predicted/confusion-matrix figures; docs/memberC_handoff_contract.md.
Member 2 main notebook	notebooks/Member2_Full_Reproduction.ipynb; single entry point for input validation, Persian preprocessing, grouped-fold verification, five-fold model load/retrain, OOF prediction, validation/test fold ensembles, evaluation, explainability, handoff export, artifact verification and tests.
Member 2 source and configuration	src/ validation, preprocessing, fold, training, deterministic prediction, evaluation, explainability, handoff-export and verification modules; configs/model_config.json and configs/project_config.json; docs/data_contract.md, workflow.md and memberC_handoff_contract.md.
Member 2 fold checkpoints	models/fold_0 through models/fold_4, each with fine-tuned ParsBERT model weights/config/tokenizer. The large model.safetensors files may be stored outside the lightweight archive; they are required for complete new-post inference/full regeneration.
Member 2 prediction handoff	data/handoff/member2_oof_predictions_deterministic.csv; member2_validation_predictions_fold_ensemble.csv; member2_test_predictions_fold_ensemble.csv. The prediction column is the primary scalar fusion input keyed by unique_post_id.
Member 2 metrics and figures	outputs/metrics/oof_metrics.json, validation_metrics.json and test_metrics.json; outputs/figures/scatter_validation.png, scatter_test.png, confusion_matrix_validation.png and confusion_matrix_test.png; outputs/explainability/error_analysis_validation.json and token_attribution_examples.json.
Early scraping / historical provenance	ninisiscraper2.ipynb plus the original section-specific scrape CSVs/checkpoints; documents thread discovery, post pagination, retry/backoff, profile caching, checkpoint recovery, and robust post-ID/content deduplication.
Finalized early cleaning notebook	02_Cleaning_Reproducible.ipynb (historical-faithful real-data run): exact 414,994 -> 413,180 -> 362,026 -> 362,017 checkpoints, raw-content preservation, category recovery, historical gender outcome replay, canonical IDs, audits, and final cleaned-data contract.
Finalized stress-proxy notebook	03_Stress_Proxy_and_Label_Sampling.ipynb: seed/surface features, initial proxy, differential-stem induction, refined proxy, historical-reference diagnostics, real-corpus distribution figures, and the reconstructed 500/600/900 annotation-sampling
Legacy cleaning/proxy provenance	Historical source notebooks (cleaner.ipynb, cleaner2.ipynb, cleaner3.ipynb) informed the reconstruction and are documented in Part 1's LEGACY_MAPPING.md; they are not duplicated in the compact repository package.

Member C notebooks executed successfully without errors and reproduced the locked-test

artifacts. The Member 1 final package contains its five native CatBoost fold models and reproducible OOF/holdout handoffs. The Member 2 final package reproduces the frozen OOF/validation/test handoff files, metrics, figures, contracts and notebook workflow; full Transformer regeneration/new-post inference additionally requires the five large fine-tuned fold checkpoints, which may be retained separately from the compact archive. All final branch and fusion joins use `unique_post_id`.